

NetCfg: A Declarative Network Configuration Compiler

Formal Methods in Configuration Synthesis

Simon Knight, Ph.D.

March 2026 • Version 2.3.0

Last updated: March 11, 2026



Abstract. NETCFG is a Rust-based network configuration compiler that takes a richly annotated topology—the “source code” of the network—and compiles it into verified, target-specific device configurations. A companion blueprint selects which compilation strategies to apply. The compiler processes these inputs through a five-layer pipeline: (1) Intent Transformation, (2) Topological Validation, (3) Lowering to a vendor-neutral Device IR, (4) Target-Specific AST Transformation, and (5) Syntax Serialisation. This report documents the data model, the transformation DSLs, and the AST-based lowering mechanism that decouples architectural logic from physical hardware constraints.

Contents

1	Introduction	2
2	Conceptual Model	3
2.1	A mental model of the compiler	3
2.2	The network model as a layered graph	4
2.3	Three input documents	6
2.4	Topology provenance	9
2.5	Blueprints compose the model declaratively	9
2.6	Mini worked example (conceptual)	9
2.7	The select-enrich-validate pattern	10
3	Background: Network Configuration as Code	11
3.1	Layers	11
3.2	Per-node state: DeviceContext	11
4	Intent and Policy Expression	12
4.1	The five concerns of a network blueprint	13
4.2	From plan to network model: declarative composition	13
4.3	Expressing policy at the plan layer	14
4.4	Reusable design-rule libraries	14
5	Blueprint DSL: Declarative Topology Transformations	15
5.1	Schema and layer ordering	15
5.2	Selector DSL (CEL surface syntax)	15
5.3	Named groups	16
5.4	Named objects	17
5.5	Primitive catalogue	17
5.6	Design-rule assertions	26
5.7	Named rules (CEL macros)	27
5.8	Blast radius policies	28
5.9	Target-specific transforms	28
6	Compilation Pipeline	29
6.1	Layer 1: Intent Transformation	29
6.2	Layer 2: Topological Assertions	29
6.3	Layer 3: DeviceIR and the Mapping DSL	30
6.4	Layer 4: Target-Specific Transform (TSDM)	30
6.5	Layer 5: Syntax Serialisation	31
7	Core Mechanisms	32
7.1	Five-pass IPAM (provision_ips)	32
7.2	Protocol overlay construction (build_protocol_layer)	33
7.3	Configuration diffing (DiffEngine)	34
7.4	AST Transformations (TSDM Layer)	35
8	CLI Reference	36
8.1	netcfg generate	37
8.2	netcfg validate	37
8.3	netcfg plan	37
8.4	netcfg inspect	38
8.5	netcfg ci-run	38
8.6	netcfg import	38
8.7	netcfg lint	38
8.8	netcfg fmt	39
8.9	netcfg impact	39
8.10	netcfg report	39
8.11	netcfg export-clab	39
8.12	netcfg sync	39
9	End-to-End Worked Example	40
9.1	Input: Blueprint	40
9.2	Stage 1: Parse and validate	41
9.3	Stage 2: Shadow topology	41

9.4	Stage 3: Primitive execution	41
9.5	Stage 4: Mapping evaluation.	41
9.6	Stage 5: Render	41
9.7	Stage 6: Atomic write	42
9.8	Case Study: Compliance-Hardened Financial Network.	42
10	Error Model	43
11	Editor and Service Integration	44
11.1	Language Server (LSP)	44
11.2	REST API Microservice.	44
12	Evaluation	44
12.1	Case study corpus	44
12.2	Abstraction ratio	45
12.3	Performance	45
13	Limitations	46
13.1	edge_properties not applied (mesh_nodes)	46
13.2	Mapping AST inspection not implemented	46
13.3	Shadow validation is a placeholder	46
13.4	No intra-primitive rollback.	46
13.5	generate CLI bypasses mapping document	46
13.6	No incremental compilation	46
13.7	No runtime verification or drift detection	46
13.8	Shadow topology cloning cost	47
13.9	WASM plugin is experimental	47
13.10	Mapping document limitations.	47
13.11	NSoT push not implemented (netcfg sync push).	47
14	Related Work	47
15	Future Work	48
A	Blueprint YAML Schema Reference	49
B	Selector DSL Grammar	55
C	CLI Quick Reference	56
	List of Figures	56
	List of Tables	58
	List of Design Decisions & Rules	60
	Revision History	60

1 Introduction

The configuration of large-scale network infrastructure is essentially a compilation problem.

Shenker [1] diagnosed the fundamental issue: unlike virtually every other discipline in computer science, networking has addressed complexity by adding protocols rather than creating abstractions. The result is a “bag of protocols”—each solving one narrow problem, with no composable interfaces between them. He proposed three abstractions for software-defined networking—a forwarding interface, a global network view, and a specification language—that would decouple intent from mechanism.

Gill et al. [2] quantified the operational cost of this absence: analysing five years of failure data from Microsoft’s data centres, they found that misconfiguration accounted for the largest share of customer-impacting incidents, with operator error during maintenance being the dominant trigger. As data-centre networks have since grown to millions of lines of device configuration, the case for automated, intent-driven synthesis has only strengthened—yet most automation remains bespoke and not transferable.

More recently, Google’s MALT [3] system introduced *multi-abstraction-layer topology* modelling, demonstrating that a shared topology representation spanning physical, logical, and design layers could unify over 100 internal teams. MALT’s key insight—that “almost all other network-management data either derives from its topology, constrains how to use a topology, or associates resources with specific places”—directly motivates NETCFG’s graph-centric design.

Meanwhile, model-driven management via OpenConfig [4] and YANG [5] has standardised vendor-neutral *per-device* data schemas, but these models do not capture cross-device topology relationships or support graph-wide operations like IP allocation across edges.

: Hardware Independence

Architectural logic must be lowerable to a vendor-neutral Intermediate Representation (DeviceIR) before any target-specific syntax is generated. This ensures that policy intent remains decoupled from physical ASIC constraints.

What NETCFG is

NETCFG is a command-line tool and Rust library that compiles an *annotated topology* into per-device configuration files. The annotated topology is the primary input: a graph of devices and links enriched with semantic properties—roles, sites, address pools, protocol flags, trust levels—that encode the operator’s intent at a network-wide level. A companion *blueprint* selects which compilation strategies (meshing patterns, allocation policies, assertion rules) to apply when deriving device-level state from those annotations. Optionally, a *mapping* document describes how to extract vendor-specific configuration stanzas from the enriched topology. The compiler produces deterministic, diffable, auditable configuration artefacts for every device in the network.

NETCFG replaces the manual “glue” layer between source-of-truth systems (NetBox [6], Nautobot [7]), automation frameworks (Ansible [8], Nornir [9]), and network modelling tools (Batfish [10]). It treats network configuration as a *compilation problem*: topology intent is compiled through a typed pipeline, not interpolated through ad-hoc templates. Where OpenConfig standardises the *schema* of device state, NETCFG standardises the *derivation* of that state from topology intent. Where MALT provides a shared topology *representation*, NETCFG provides a topology *compiler* that transforms intent into per-device configuration.

Design philosophy

Three principles explain every design decision in this document:

1. **Lowering over templating.** The operator’s intent is lowered through successive stages of abstraction, mirroring the multi-stage lowering architecture of LLVM [11]. Just as LLVM lowers high-level source through a language-independent IR (LLVM IR) to target-specific machine code, NETCFG lowers topology intent through a vendor-neutral DeviceIR to platform-specific CLI syntax. The analogy is precise: LLVM’s frontend/IR/backend separation corresponds to NETCFG’s blueprint/DeviceIR/template separation. This ensures that hardware quirks and vendor-specific logic are handled via structured AST transformations, not ad-hoc string concatenation.
2. **Separation of what from how.** The annotated topology declares *what* the network contains—devices, links, and their semantic properties. The blueprint declares *which* design rules to apply when compiling that topology (meshing style, allocation strategy, assertion constraints). The mapping and templates declare *how* to render the enriched model into per-device configuration. Platform-specific rules are injected via imports: `[ 'hardware-lowering.yaml' ]` and isolated behind when: `device_os == ... predicates`. Thus, the same topology and blueprint can produce Cisco IOS, Arista EOS, or Juniper JunOS output by swapping only the mapping and templates.

Insight:
Manual CLI entry is the “assembly language” of networking; NETCFG provides the high-level language and compiler necessary for scale.

3. **Topology as the single source of truth.** Following MALT’s insight that topology is the root from which all other network-management data derives, `NETCFG` treats the graph as the canonical representation. IP addresses, protocol sessions, policy rules, and security zones are all *computed* from topology relationships, not declared independently.

Audience

This document is intended for:

- Engineers integrating `NETCFG` into a network automation pipeline.
- Contributors to the `NETCFG` and `NTE` codebases.
- Researchers evaluating graph-based approaches to network configuration.

Familiarity with YAML, basic graph theory, and IP networking is assumed. Experience with Rust is helpful but not required for the API sections.

How to read this document

§2 introduces the conceptual model—the layered graph, blueprint composition, and the select/enrich/validate pattern—without any code or formalism. §3 then grounds these concepts in the formal data model and Rust implementation. Readers wanting the blueprint syntax should continue to §5 (blueprint DSL) and §8 (CLI reference), referring to individual sections as needed. §6–§7 cover the compilation pipeline and core mechanisms in depth and can be deferred until required. §9 provides a complete end-to-end worked example that traces a blueprint through every pipeline stage. §11 documents the language server and REST API for editor and CI/CD integration.

2 Conceptual Model

Before introducing formal definitions or code, this section explains the three ideas that underpin every `NETCFG` blueprint: (1) the network model is a *layered graph*; (2) blueprints *compose* the model declaratively, layer by layer; (3) every primitive follows the same *select* → *enrich* → *validate* pattern. Understanding these concepts makes the formal data model (§3) and DSL syntax (§5) easier to absorb.

Keep the mental model, postpone the mechanisms. In this section, treat `NETCFG` as a compiler operating over a graph: selectors identify where in the graph to act, primitives add derived state, and assertions act as quality gates. The implementation details (Rust structs, CEL compilation, DataFrames) exist to make this model fast and safe, but they are not required to understand the flow.

2.1 A mental model of the compiler

At a high level, `NETCFG` takes a richly annotated topology and progressively derives device-level state from its annotations until each node contains enough information to render deterministic, diffable configuration. Figure 1 shows the compilation funnel; Table 1 summarises the artefact produced at each stage.

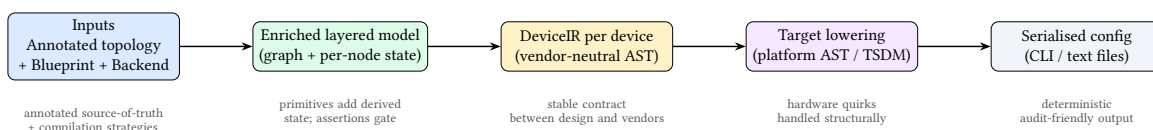


Figure 1. Conceptual compilation funnel. The primary input is the annotated topology; the blueprint selects compilation strategies; the compiler derives an enriched layered graph; the backend lowers that graph into vendor-neutral and then target-specific ASTs before serialisation.

Table 1. Stage outputs (conceptual artefacts). These conceptual stages map to the five pipeline layers in §6: Compose = Layer 1, Lower = Layer 3, Adapt = Layer 4, Serialise = Layer 5.

Stage	Output artefact	What it represents
Input	Annotated topology + blueprint + backend package	Richly annotated source-of-truth graph, compilation strategy directives, and a reusable target backend.
Compose	Enriched layered model	A single graph containing physical and logical layers, with per-node derived state.
Lower	DeviceIR	Vendor-neutral, structured representation of per-device intent.
Adapt	Target AST / TSDM	Platform-specific structure (interface naming, feature support, syntax constraints).
Serialise	Text configuration files	Deterministic device configs suitable for diff, review, and deployment.

2.2 The network model as a layered graph

NETCFG represents the network as a graph of *nodes* (devices, protocol instances) and *edges* (links, peering sessions). The graph is organised into *layers*: the *physical layer* contains routers, switches, and cables; each *protocol layer* (BGP, OSPF, etc.) contains a parallel set of nodes—one per device participating in that protocol—linked back to their physical origin.

What a “layer” means. A layer is a named, self-contained subgraph representing one plane of intent. The physical layer captures what exists as hardware and cabling. Protocol layers capture logical relationships (adjacencies, sessions, overlay bindings) that are *hosted* on those physical devices. Layers are not “views” over the same nodes; each layer has its own node set. The compiler can therefore represent, in one model, both “device A is cabled to device B” (physical) and “A peers with C via a route-reflector” (BGP) even when those relationships do not coincide.

A physical router **A** thus has a corresponding BGP node **A’** and (optionally) an OSPF node **A”**. Each protocol node carries a *parent mapping* that traces it back to the physical device that hosts it. This cross-layer traceability is how IP addresses allocated on the physical layer flow into BGP configuration without explicit wiring.

Parent mapping and traceability. Every non-physical node stores the identifier of its host physical device (`_source_id` in `data_json`). This forms a many-to-one mapping from overlay nodes to physical nodes: multiple protocol instances (BGP, OSPF, EVPN, IPsec) may all point back to the same physical router. The parent mapping is the only cross-layer link.

Figure 2 illustrates a minimal example. The physical layer shows three interconnected devices. The protocol layer above contains corresponding protocol nodes; dashed arrows show the parent mapping.

Figure 3 extends the same idea to multiple overlays. Each physical device can host many protocol-layer nodes, one per protocol.

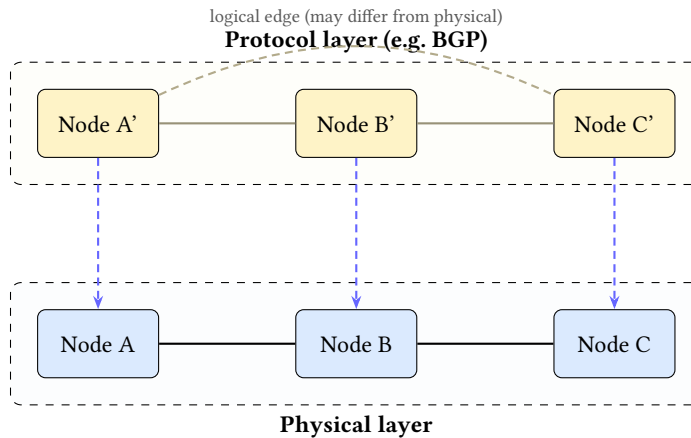


Figure 2. The layered graph model. Physical devices (bottom) have corresponding protocol nodes (top). Dashed arrows show the parent mapping (`_source_id`) that traces each protocol node back to its physical origin. Protocol-layer edges represent logical intent relationships and may be derived from, but need not mirror, physical links.

Insight:
Because overlays never directly reference other layers, the compiler can clone, transform, and validate each layer independently while preserving an unbroken “why” trail back to the physical topology.

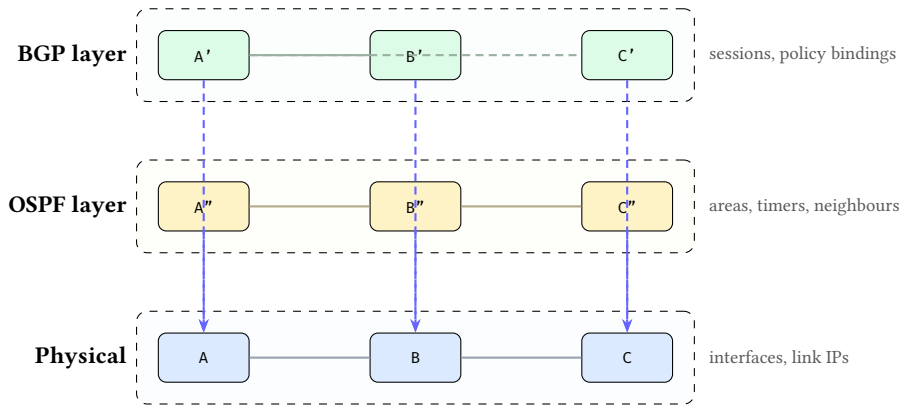


Figure 3. Layered stack as bands. Each column is a physical device (A, B, C). Each protocol layer adds one node per participating device, linked back to the physical node via the parent mapping (vertical dashed arrows). Overlay edges encode protocol-specific logical relationships and may differ across layers.

Where facts live (and how they flow across layers). As a rule of thumb, facts originate in the layer where they are naturally defined: (1) *link and interface facts* (peer interfaces, link subnets, L2/L3 addressing) originate on the physical device nodes; (2) *protocol intent* (neighbour sets, areas, timers, policy bindings) originates on the protocol-layer nodes.

When a protocol node needs a physical fact (e.g., a neighbour IP), it obtains it by following its parent mapping to the host physical node and reading the already-derived fields there. This is how “allocate on physical, consume in overlay” works without bespoke wiring.

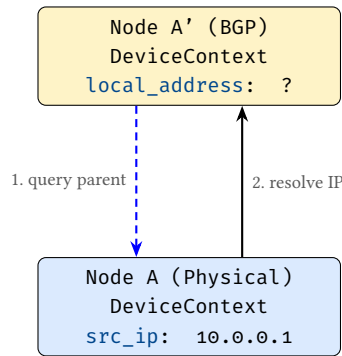


Figure 4. Cross-layer state resolution. Protocol overlays do not maintain redundant state. They resolve physical facts (like allocated IPs) on-the-fly by querying their host physical node via the `_source_id` parent mapping.

Edges within a layer represent direct relationships: physical cables, peering sessions, or policy bindings. Edges *never* cross layers; the parent mapping is the sole mechanism for cross-layer traceability. All mutable state—IP addresses, protocol parameters, security zones—lives on *nodes*, not on edges. Edges exist solely to express relationships between nodes.

Why node-centric state matters. Configuration is ultimately rendered per device. By ensuring every piece of renderable information is attached to nodes (and by keeping edges as pure relationships), the mapping stage can generate a device’s configuration by reading exactly one node’s accumulated DeviceContext. Edges still drive computation (e.g., “allocate a /30 per link”), but the results of that computation are always written back onto the endpoint nodes.

Layering as a compilation mechanism. Layers are created and populated by primitives. For example: `mesh_nodes` adds physical edges; `provision_ips` annotates the physical nodes with link addresses; `build_protocol_layer` clones a selected set of physical nodes into a protocol layer and deep-merges a protocol overlay into the clones. Subsequent primitives can then select either physical nodes or protocol nodes by `layer==....`

Note

Design note: protocol edges are logical, not physical. Protocol-layer edges represent intent-level relationships (e.g., “BGP session exists between A’ and C” or “policy binds to this adjacency”). They may be derived from physical links, but they are not required to mirror the physical layer’s connectivity.

Kinds of layers (a practical taxonomy)

Layers are intentionally lightweight: they let the model hold multiple simultaneous views of the same network without conflating their semantics. In practice, most `NETCFG` blueprints use a small, repeatable taxonomy:

1. **Physical layer.** Nodes represent devices; edges represent physical links. Typical state: interface naming, link addressing, management reachability, hardware inventory.
2. **Protocol layers (routing/switching overlays).** Nodes represent a device’s participation in a protocol (one node per device per protocol); edges represent logical adjacencies or sessions. Typical state: neighbours, timers, areas/ASNs, policy attachments, protocol feature flags.
3. **Policy and service layers (optional).** Some organisations model policy and services as dedicated layers when the relationships are graph-shaped (e.g., security-zone bindings, NAT relationships, telemetry subscriptions). Others keep these as node metadata attached to the physical or protocol nodes. The design rule is the same either way: *renderable state stays on nodes; edges express relationships*.

This taxonomy is not prescriptive: the point is that layering preserves separation of concerns while keeping every derived fact traceable back to the physical topology via the parent mapping.

Design rules that build overlays

Layering alone provides representation; *design rules* provide synthesis. In `NETCFG`, design rules are implemented as primitives that read from the current model (often the physical layer) and then materialise overlay structure and per-device stanzas. They are the bridge from plan topology to protocol overlays.

Two common classes of design rule are:

1. **Overlay construction rules.** `build_protocol_layer` clones selected physical nodes into a protocol layer, records the parent mapping (`_source_id`), and deep-merges protocol defaults into the clones. With `clone_underlying: true`, it also seeds the overlay with a derived copy of the physical connectivity.
2. **Session synthesis rules.** Protocol session primitives (e.g. `build_bgp_session`, `build_ospf_session`, `build_isis_session`)¹ consume the overlay graph and emit configuration stanzas on the participating nodes. These rules are where operator intent becomes concrete neighbours, interfaces, and per-peer parameters.

2.3 Three input documents

An operator supplies three documents to the compiler:

1. **Annotated topology** — the “source code” of the network. A graph of devices (nodes) and links (edges), where each node carries rich semantic annotations: `hostname`, `role`, `site`, `device_os`, trust levels, address pools, protocol flags, and any other properties the operator considers part of the network’s design intent. The topology may be hand-authored (YAML/JSON), imported from a network inventory system via `netcfg import` (§8.6), or pulled live from an NSoT such as NetBox [6] via `netcfg sync pull` (§8.12). These annotations are the data that the compiler’s selectors (e.g. `nodes[role=='core']`) and primitives operate on. The richer the annotations, the more the compiler can derive automatically. In the ideal case the annotated topology captures *all* network-wide design intent; the blueprint then reduces to a thin set of compilation strategy choices.
2. **Blueprint** — *which* compilation strategies to apply. A YAML document declaring layers of primitives that select subsets of the annotated topology and derive new state: meshing patterns, IP allocation pools and strategies, protocol overlay construction, policy rules, and assertion constraints. The blueprint does not duplicate what is already in the topology; rather, it names the *design rules* that transform topology-level annotations into device-level state.

The relationship is analogous to a traditional compiler: the annotated topology is *source code* and the blueprint is a *compilation profile* (flags such as `-O2` or target architecture). The source code carries the program’s semantics; the compilation profile selects *how* those semantics are lowered. Similarly, the topology carries network intent; the blueprint selects which primitives lower that intent into device-level state.

¹Session synthesis primitives are planned but not yet in the catalogue; the current workflow uses `build_protocol_layer` with a config overlay to achieve the same result.

Insight:

The operator does not “mesh in the BGP layer” manually. They declare intent; the design rules materialise the overlay relationships and the resulting per-device stanzas in the appropriate layer.

3. **Target backend** — *how* to turn the enriched model into per-device configuration. In practice this is a small directory containing (i) target-specific transforms that lower vendor-neutral intent into platform-specific structure, and (ii) templates that serialise that structure into concrete CLI or configuration syntax. Most organisations author this once per platform and reuse it across sites.

Note

Where does intent live? The boundary between topology annotations and blueprint declarations is a design choice that operators can tune. At one extreme, the topology carries only physical inventory (hostnames and cabling) and the blueprint encodes all design rules. At the other, the topology carries rich intent (address pools, protocol parameters, zone memberships) and the blueprint is a thin strategy selector. NETCFG supports both styles; the general trajectory is toward richer topologies with thinner blueprints, because this keeps intent closer to the source-of-truth and reduces the coupling between the blueprint and a specific topology's node schema.

2.3.1 Canonical annotation vocabulary

Table 2 lists the well-known annotation keys that NETCFG recognises on topology nodes. Operators may attach *any* key-value pair; the keys below receive special treatment from selectors, primitives, or the rendering engine.

Table 2. Well-known topology annotation keys. Keys marked with * are computed by the selector engine and need not appear in the topology file. All other keys are authored by the operator (or imported via `netcfg sync pull`).

Key	Type	Consumed by
hostname	string	All primitives (node identity)
role	string	Selectors (<code>nodes[role=='spine']</code>)
site	string	Selectors, multi-site blueprints
device_os	string	Template selection, vendor transforms, QoS constraints
management_ip	string	Inventory; passed through to templates
zone	string	<code>build_zone_policy</code> , zone-based assertions
platform	string	Hardware inventory transforms
vrf	string	<code>provision_ips</code> (routing context separation)
disabled	bool	Selectors (exclude nodes from processing)
<i>Computed selector variables *</i>		
id	integer	Node identifier in the graph
layer	string	Current graph layer (e.g. physical, bgp)
degree	integer	Number of adjacent edges
is_leaf	bool	True when <code>degree == 1</code>
neighbors	list	Adjacent node identifiers

Any key not listed above is stored in the node's metadata map and remains accessible to selectors and templates. For example, an operator might annotate nodes with `trust_level: high` or `loopback_pool: 10.255.0.0/24`; these values are available to CEL selectors and MiniJinja templates without any schema changes.

2.3.2 Thin blueprint vs. rich blueprint

The following side-by-side example shows the same three-router network expressed in two styles. Both produce identical compiled output.

(a) Bare topology + fat blueprint

```
# topology.yaml -- minimal
nodes:
- id: r1
  type: Router
- id: r2
  type: Router
- id: r3
  type: Router

# blueprint.yaml -- all intent here
layers:
- name: physical
  primitives:
    - type: mesh_nodes
      selector: "nodes[type=='Router']"
      mesh_type: full
    - type: provision_ips
      selector: "edges[true]"
      pool: 10.0.0.0/24
      subnet_size: 30
    - type: tag_nodes
      selector: "nodes[id=='r1']"
      tags: {role: core, site: dc1}
    - type: tag_nodes
      selector: "nodes[id=='r2']"
      tags: {role: edge, site: dc1}
```

(b) Rich topology + thin blueprint

```
# topology.yaml -- carries intent
nodes:
- id: r1
  type: Router
  properties:
    role: core
    site: dc1
    device_os: eos
- id: r2
  type: Router
  properties:
    role: edge
    site: dc1
    device_os: eos
- id: r3
  type: Router
  properties:
    role: edge
    site: dc2
    device_os: junos

# blueprint.yaml -- strategy only
layers:
- name: physical
  primitives:
    - type: mesh_nodes
      selector: "nodes[role != '']"
      mesh_type: full
    - type: provision_ips
      selector: "edges[true]"
      pool: 10.0.0.0/24
      subnet_size: 30
```

In style (a), the blueprint must name individual nodes and inject properties via `tag_nodes`—tightly coupling the blueprint to this specific topology. In style (b), the topology carries its own identity (roles, sites, platforms), and the blueprint operates on *semantic groups* (`nodes[role != " ]`). The same blueprint works unchanged if a fourth router is added to the topology.

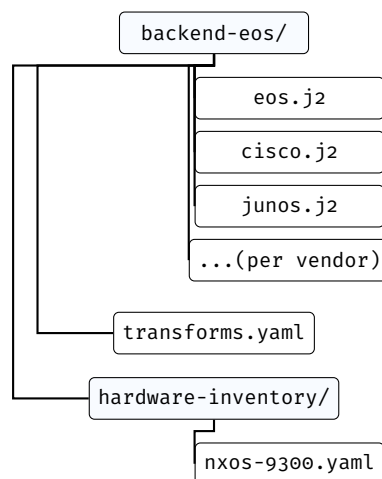


Figure 5. Target backend package structure. Templates render per-vendor syntax; `transforms.yaml` declares TSDM lowering rules; `hardware-inventory/` maps chassis models to slot layouts. Most organisations author this once per platform and reuse it across sites.

The annotated topology is the primary authoring surface: it captures the network’s identity, structure, and design intent. The blueprint selects compilation strategies; the mapping document and templates are typically authored once per vendor platform and reused across sites.

2.4 Topology provenance

The annotated topology can enter NETCFG through three channels, each preserving the node annotations that selectors and primitives operate on:

1. **Hand-authored YAML/JSON.** The operator writes the topology file directly, placing each node's annotations under a `properties:` block. This is the most explicit channel and is common during greenfield design or lab work.
2. **Import from existing configurations** (`netcfg import`, §8.6). Parses a directory of device configurations (or a device inventory file) and produces a topology archive (`.nte`), inferring hostnames, links, and device platforms from the configuration data. Operators then enrich the imported topology with additional annotations (roles, sites, trust levels) before compiling.
3. **NSoT synchronisation** (`netcfg sync pull`, §8.12). Pulls a live topology from a network source of truth such as NetBox [6] via its GraphQL API, mapping inventory fields to topology annotations. This channel keeps the annotated topology in sync with the production network.

Regardless of channel, the result is the same data structure: a directed graph with a JSON property blob per node. Primitives and selectors are agnostic to provenance—a topology imported from NetBox is compiled identically to one hand-authored in YAML.

2.5 Blueprints compose the model declaratively

A blueprint organises primitives into ordered *layers*. Each layer declares a set of primitives; each primitive selects a subset of the graph and enriches the nodes it touches. The result is a progressively richer model, as Figure 6 illustrates.

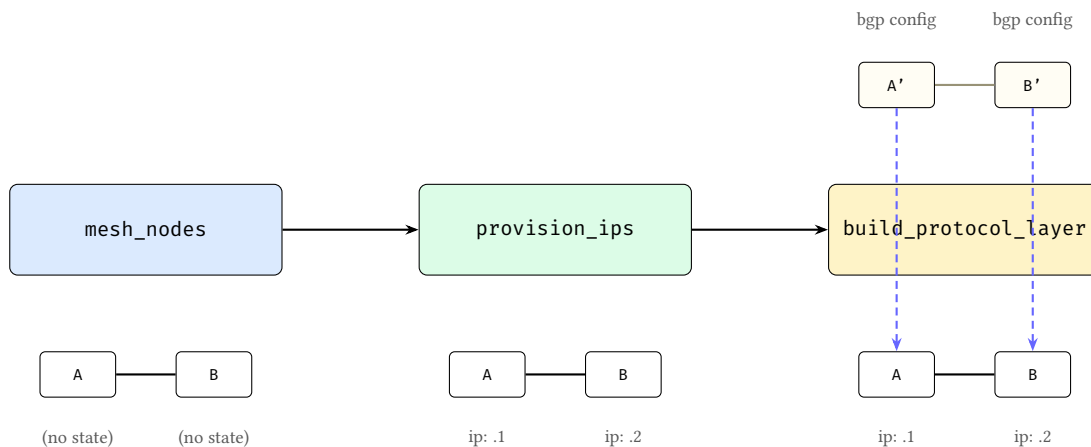


Figure 6. Progressive model enrichment. Left: `mesh_nodes` establishes edges between bare nodes. Centre: `provision_ips` annotates nodes with IP addresses. Right: `build_protocol_layer` clones nodes into a protocol layer and deep-merges a protocol overlay into the clones; dashed arrows show the parent mapping back to the physical devices. At each stage the model grows richer; no primitive needs to know what others have done.

Three properties make this composition model work:

1. **State lives on nodes, not edges.** Edges define relationships—physical links, peering sessions, policy bindings. All mutable, renderable state is written to the node at each end of an edge. Edges may carry static relationship metadata (e.g., labels), but do not carry configuration-bearing state.
2. **Primitives compose, not override.** Each primitive enriches the model additively. No primitive needs to know what other primitives have done; the model accumulates state layer by layer.
3. **Primitives are typed model transforms.** Each primitive produces a specific kind of enrichment: new graph structure (overlay nodes and edges), allocated identifiers (IP addresses, ASNs, VLANs), compiled policy objects (firewall rules, route-maps), or protocol stanzas (VXLAN VNIs, EVPN route-targets). Primitives never name specific devices; they reference *groups* and *selectors* (graph queries that resolve against the current model state), so the same blueprint can operate on topologies of different sizes.

2.6 Mini worked example (conceptual)

The following tiny blueprint fragment illustrates the layered-model flow without requiring any knowledge of Rust, template engines, or selector internals.

Listing 1. A minimal blueprint fragment (layered model + design rules).

```

layers:
  - name: physical
    primitives:
      - type: mesh_nodes
        selector: "nodes[role=='core']"
        mesh_type: full

      - type: provision_ips
        selector: "edges[true]"
        pool: 10.0.0.0/24
        subnet_size: 30

  - name: bgp
    requires: [physical]
    primitives:
      - type: build_protocol_layer
        selector: "nodes[layer=='physical' && role=='core']"
        layer: bgp
        clone_underlying: true
        config: { bgp: { enabled: true } }

    # With clone_underlying: true, BGP overlay edges are derived
    # automatically from the physical topology. No separate
    # session primitive is needed.

assertions:
  - name: core-has-link-address
    severity: error
    select: "nodes[role=='core']"
    check: { type: field_exists, field: src_ip }

```

Conceptually, this executes as follows:

1. **Start with an annotated topology** containing core routers as physical nodes.
2. **mesh_nodes** creates physical edges between those nodes.
3. **provision_ips** examines the edges to decide which links need subnets, then writes the resulting link addresses onto the endpoint *nodes* (not onto the edges).
4. **build_protocol_layer** clones the physical core nodes into a BGP overlay layer and records a parent mapping (`_source_id`) back to the physical devices. The protocol overlay config is deep-merged into the clones.
5. **Overlay edges encode protocol intent.** With `clone_underlying: true`, the BGP overlay begins with derived connectivity copied from the physical layer. The compiler then reads that overlay graph and writes per-device neighbour stanzas onto the participating nodes.
6. **Assertions** verify the model is well-formed (here: every core device has a link address) before any vendor-specific configuration is produced.

The key point is that the operator never names a specific interface or assigns an IP address directly. The blueprint describes intent, and the compiler derives concrete per-device state from the layered topology relationships.

2.7 The select–enrich–validate pattern

Every primitive follows the same three-phase pattern (Figure 7):

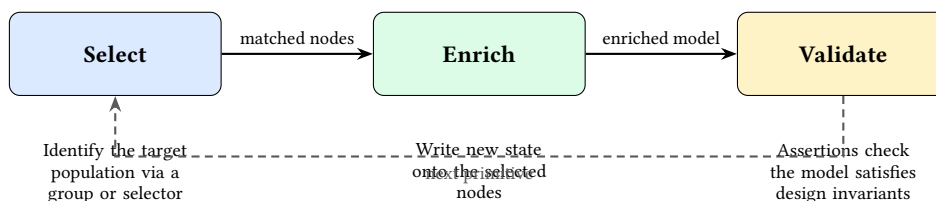


Figure 7. The select–enrich–validate cycle. Every primitive selects a population of nodes or edges and enriches them with new state. Validation is expressed as assertions over the model and may run as an explicit checkpoint, a layer barrier, or a final gate before rendering.

- **Select.** A group reference or CEL selector query identifies which nodes (or edges) the primitive will operate on. The selector is evaluated against the current graph state, so earlier primitives’ enrichments are visible to later selectors.
- **Enrich.** The primitive performs a typed model mutation on the selected population. Depending on the primitive’s tier, this may produce new graph structure (overlay nodes and edges), allocated identifiers (IP addresses, ASNs, VLAN IDs), compiled policy objects (firewall rules, route-maps, prefix-lists), or protocol stanzas (VXLAN VNIs, EVPN route-targets, BGP peering parameters). State always accumulates; primitives never delete or overwrite each other’s contributions.
- **Validate.** Assertions verify that the model satisfies the operator’s invariants (e.g., “every node has a link address” or “no two links share a subnet”). Assertions may run immediately (as checkpoints), at the end of a layer, and again as a final gate before any configuration is generated.

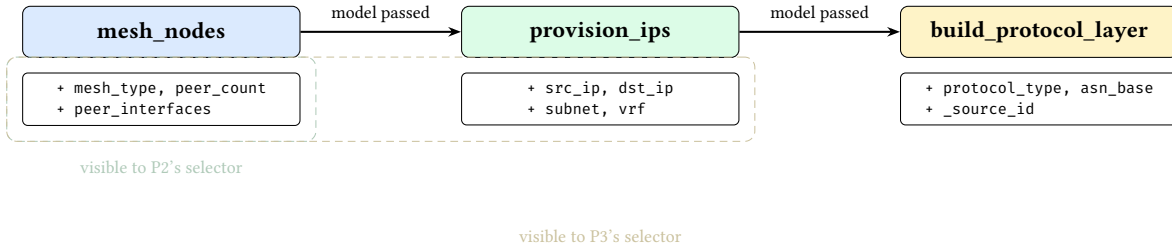


Figure 8. Selector visibility timeline. Each primitive’s selector evaluates against the *current* graph state, which includes all fields written by earlier primitives. State accumulates monotonically.

Validation checkpoints. NETCFG supports three complementary validation styles, all expressed as assertions over the current model state: (1) *mid-pipeline checkpoints* (via `assert_state`) that fail fast after a critical enrichment step; (2) *layer barriers* that run after a layer’s primitives execute; and (3) *final gates* that run before any configuration is rendered. Conceptually, all three ask the same question—“does the model satisfy the design invariants?”—but they fire at different times for faster feedback.

This uniform pattern means that learning one primitive teaches the operator the shape of every primitive: declare a selector, declare the desired state, and optionally assert invariants. The formal definitions and concrete DSL syntax follow in §3 and §5.

3 Background: Network Configuration as Code

Concept first. The purpose of the formal model in this section is to make the compiler’s behaviour precise: what constitutes a node, what constitutes an edge, how layers coexist, and how per-device state accumulates. Implementation choices (Rust types, query engines, evaluation languages) are replaceable details; the stable ideas are: a layered graph, node-centric state, and deterministic derivation of device configuration from topology relationships.

NETCFG models the network as a directed graph $G = (V, E, \tau, \lambda, D)$ where V is a set of nodes (devices, protocol instances), $E \subseteq V \times V$ is a set of edges (links, peering sessions), $\tau : V \rightarrow T$ assigns each node a *type* (Router, Switch), $\lambda : V \rightarrow L$ assigns each node a *layer*, and $D : T \rightarrow \text{DataFrame}$ maps each type to a columnar property store (Polars DataFrame).

3.1 Layers

A layer represents a plane of the network: `physical` for the hardware topology, `bgp` for BGP peering sessions, `ospf` for OSPF adjacencies. A single physical router (node 1, layer `physical`) may have corresponding protocol nodes (node 101, layer `bgp`; node 201, layer `ospf`), each linked to node 1 via a *parent mapping* (`_source_id` in the node’s `data_json`).

Layers allow the compiler to model the physical and logical topologies simultaneously, with cross-layer references preserving traceability from a protocol node back to the physical device that hosts it.

3.2 Per-node state: DeviceContext

How per-node state relates to the layered model. The layered graph defines *where* intent lives: a device’s physical node captures hardware-adjacent facts (interfaces, link addressing, management), while each protocol-layer node captures logical intent for that protocol (BGP neighbours, OSPF areas, EVPN VNIs). In both cases, the compiler stores derived configuration-bearing state *on the node* so that rendering is a local operation: to generate configuration for a device in a given layer, the backend reads exactly one node’s accumulated state.

Conceptually: a node’s “accumulated facts”. Think of DeviceContext as a typed view over a node’s accumulated facts at that point in the pipeline. Primitives add facts (“this interface has 10.0.0.1/30”, “this device is in VRF X”, “these are my BGP peers”), and assertions check that required facts are present before lowering continues.

Each node’s properties are stored as a JSON blob in a Polars DataFrame keyed by node type. Polars provides ultra-fast topological graph queries and edge traversals, unpacking the JSON payload on-demand only when evaluating CEL [12] selector expressions. The compiler maintains a per-node *device context* whose fields are progressively enriched by primitives:

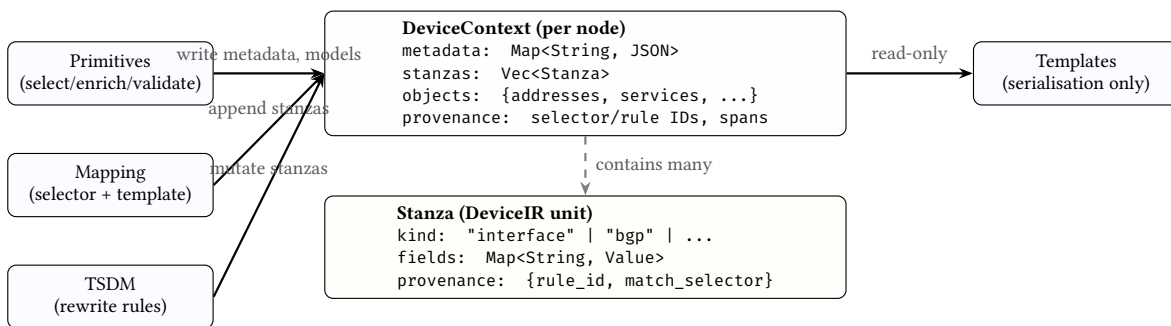
Table 3. Per-node device context: fields populated by the compiler pipeline. Fields marked with † are written by specific primitives; all others originate from the annotated topology.

Field	Type	Source
hostname	string	Topology node id
device_os	string?	Topology annotation
management_ip	string?	Topology annotation
src_ip, dst_ip, subnet	string?	Written by provision_ips †
src_ip6, dst_ip6, subnet_v6	string?	Written by provision_ips † (dual-stack)
vrf	string?	Topology annotation or allocate_resources †
peer_interfaces	map	Written by mesh_nodes †
stanzas	list	Accumulated config units from all primitives †
(any other key)	JSON value	Topology annotations or primitive-written metadata

Primitives progressively enrich this context: mesh_nodes establishes edges and writes mesh_type/peer_count into metadata; provision_ips writes IP fields; build_protocol_layer deep-merges config overlays into the cloned node’s metadata. By the time rendering occurs, each node’s device context contains all information needed to produce its configuration file.

Note

Any topology annotation that does not match a named field is captured in the node’s metadata map. This makes the device context extensible without schema changes—operators can attach arbitrary properties in the topology and reference them from selectors or templates.



Key invariant: rendering reads exactly one node’s DeviceContext; no cross-node lookups at emission time.

Figure 9. DeviceContext as the compiler’s per-node state carrier. Primitives enrich metadata and typed models; mapping emits structured stanzas; TSDM rewrites those stanzas; templates then serialise the final per-node state without introducing new semantics.

Note

Graph engine dependency. The underlying graph is provided by the NTE topology engine [13], which implements a directed graph with columnar (Polars DataFrame) property storage. NETCFG uses NTE for the graph, the mapping evaluator, and DeviceIR types.

4 Intent and Policy Expression

The operator begins with a high-level picture of the network—often a whiteboard sketch or a network diagram—and formalises it into two artefacts: an annotated topology (the “source code” of the network, carrying nodes,

roles, sites, and properties) and a blueprint (the compilation strategies that derive device-level state from those annotations). The operator never manipulates IP addresses or protocol stanzas directly; the compiler derives them from the topology’s annotations via successive primitive applications.

This section frames the conceptual model that operators use when authoring these artefacts, before §5 catalogues the concrete syntax.

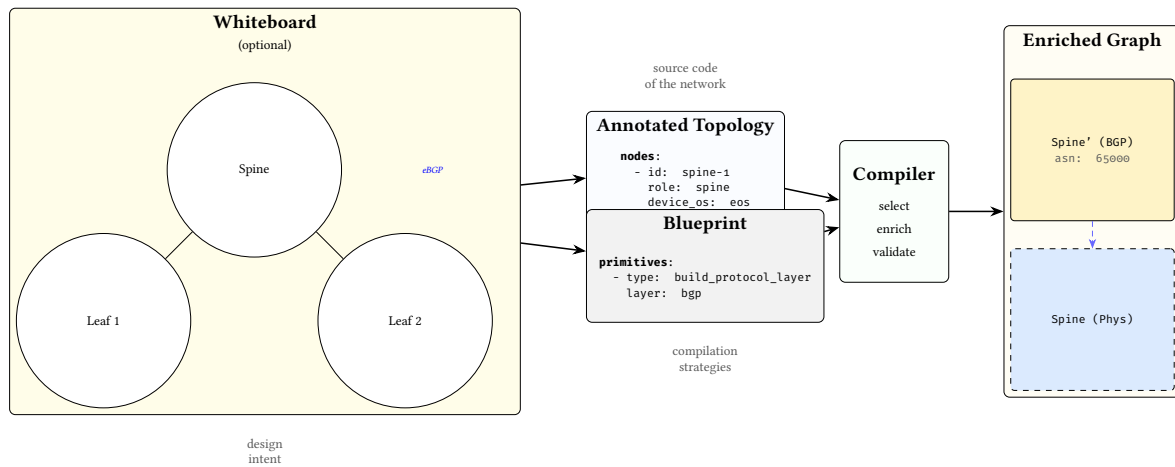


Figure 10. From whiteboard to compiled model. An architect’s visual intent (left, optional) informs both the annotated topology (the network’s “source code”: nodes, roles, sites, properties) and the blueprint (which compilation strategies to apply). Together, these feed the compiler, which produces an enriched, layered graph (right).

4.1 The five concerns of a network blueprint

Every blueprint addresses five orthogonal concerns. Table 4 shows how each maps to a DSL construct and the design question it answers.

Table 4. The five concerns of a network blueprint.

Concern	DSL construct	Question answered
Population naming	<code>groups:</code>	Which devices belong to which roles?
Topology intent	<code>layers:</code> / <code>primitives</code>	How should they be connected and what state should they carry?
Policy constraints	<code>routing/access/prefix-list</code> primitives	What traffic rules should be enforced?
Design invariants	<code>assertions:</code> + <code>rules:</code>	What properties must always hold?
Change safety	<code>blast_radius:</code>	How much change is acceptable per commit?

The key insight is that every concern operates on the *network model*—the layered graph described in §3. The operator declares populations, intent, policy, and invariants; the compiler maps these into graph mutations that progressively enrich the model’s nodes.

4.2 From plan to network model: declarative composition

§2.5 introduced the five primitive tiers and their mutation semantics. This subsection shows how those tiers interact at the implementation level.

`provision_ips` iterates edges to determine which node pairs need addresses, but writes `src_ip` and `dst_ip` onto the respective *nodes*—not onto the edge itself. `build_protocol_layer` clones selected nodes into a new graph layer and merges protocol configuration into them via deep merge. Each primitive sees the enriched model left by its predecessors, but never needs to know what they did.

: Node-Centric State

All per-device state is written to nodes via `DeviceContext`. Edges carry no mutable state; they exist solely to express relationships between nodes. This invariant simplifies the mapping stage: to produce a device’s configuration, the mapping evaluator reads exactly one node.

Groups and selectors are the glue that lets these transforms find their targets. Groups let the operator name populations once (e.g. `$core_routers`, `$edge_links`, `$management_nodes`) and reference them in every

primitive and assertion. Selectors compile group references and attribute predicates into CEL graph queries that the compiler evaluates against the live topology.

4.3 Expressing policy at the plan layer

Policy primitives follow the same declarative pattern: select a population, declare the desired policy state, and let the compiler write it to the model. The operator never constructs individual ACL entries or route-map clauses. Instead, they declare the *intent*—which traffic classes to permit or deny, which prefixes to advertise, which communities to match—and the primitive translates this into the appropriate policy objects on each selected node.

The separation between *topology intent* (layers 1–2: structure and addressing) and *policy intent* (layer 3+: traffic rules) is a deliberate design choice. Topology primitives build the graph; policy primitives annotate nodes with forwarding constraints. Both operate through the same select-and-enrich mechanism, but they address orthogonal concerns and can be developed, tested, and imported independently.

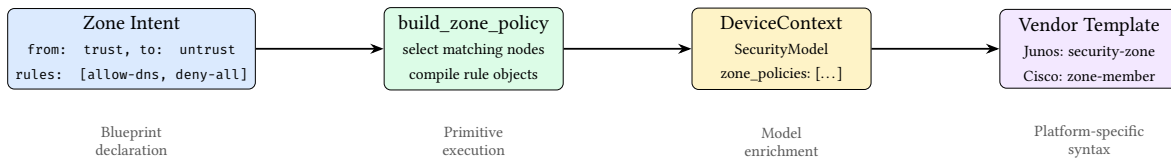


Figure 11. Policy compilation flow. The operator declares zone intent; the `build_zone_policy` primitive compiles it into typed rule objects on `DeviceContext`; vendor templates render platform-specific syntax.

Case study: multi-tenant isolation

The multi-tenant data-centre case study (`examples/case_studies/04_multi_tenant_dc.yaml`) illustrates how these concerns compose. The operator declares: (1) groups that name device roles and per-tenant link populations; (2) a hub-and-spoke mesh for the fabric; (3) per-VRF address pools; (4) access policies that deny cross-VRF traffic; (5) assertions that every device has a tenant assignment and that each tenant’s subgraph is connected; and (6) a blast-radius limit. At no point does the operator name a specific device or IP address—the compiler derives all concrete state from the declared populations and intent.

4.4 Reusable design-rule libraries

Complex assertions can be factored into named rules: (CEL macros) and shared across blueprints via `imports:`. Two standard rule libraries and a protocol fragment library ship with `NETCFG`:

- **datacenter-rules.yaml** — a portable linting ruleset for data-centre fabrics. It validates OSPF point-to-point subnet lengths, MTU consistency across edges, BGP private-ASN ranges, BGP neighbour IP existence, IS-IS level consistency, and IS-IS NET address formatting. Blueprints import it with a single line: `imports: [ 'datacenter-rules.yaml' ]`.
- **hardware-lowering.yaml** — vendor-specific AST transforms for Cisco NX-OS, Juniper Junos, Cisco IOS-XR, and VyOS. It remaps abstract Ethernet interface names into platform-specific formats (e.g. `Ethernet0` → `ge-0/0/0` on Junos) and standardises overlay MTU to 9216.
- **protocols/*.yaml** — a library of 31 importable protocol fragments, each providing a single `build_protocol_layer` invocation with sensible defaults. Covers routing (BGP, OSPF v2/v3, IS-IS, RIP, LDP, RSVP-TE, SR-MPLS), switching (VXLAN, EVPN, STP/RSTP, LACP, MLAG), security (IPsec/IKEv2, WireGuard, MACsec, 802.1X, RADIUS, TACACS+), services (NTP, Syslog, SNMP, DHCP Relay), multicast (PIM-SM, IGMP), telemetry (NetFlow/IPFIX/sFlow), and layer 2 (LLDP, ARP, BFD, VRRP, GRE). Blueprints import individual protocols: `imports: [ 'protocols/ospf.yaml', 'protocols/bfd.yaml' ]`.

Ten case studies in `examples/case_studies/` demonstrate production-scale blueprints: ISP dual-stack peering, campus compliance audit, WAN OSPF→IS-IS migration, multi-tenant DC with per-VRF isolation, SP MPLS core with SR-MPLS, campus 802.1X NAC, SD-WAN with IPsec overlay, multicast video distribution, zero-trust DC microsegmentation, and PCI-DSS financial compliance.

Assertions reference named rules via the `use_rule` check type, keeping the assertion block concise:

Listing 2. Referencing named rules via `use_rule`.

```

rules:
  link-addressed: "has(node.src_ip) && node.src_ip != ''"
  protocol-bound: "has(node.metadata.asn) && node.metadata.asn != ''"
  
```

```

assertions:
- name: every-edge-device-has-address
  severity: error
  select: nodes[role=='edge']
  check:
    type: use_rule
    name: link-addressed

```

5 Blueprint DSL: Declarative Topology Transformations

A blueprint is a YAML document declaring what topology transformations to apply. It is the “what” of the network design.

5.1 Schema and layer ordering

Listing 3. Blueprint top-level schema (YAML surface syntax). Appendix A provides the full reference.

```

version: 1 # schema version (required)
imports: [shared-rules.yaml] # merge other blueprints
layers: # ordered compilation layers
- name: physical
  requires: [] # layer dependencies
  primitives: [...] # typed model transforms
assertions: [...] # post-execution design checks
rules: # named CEL macros
  is_core: "role == 'core'"
groups: # named node sets
  core_routers: { selector: "nodes[role=='core']" }
blast_radius: [...] # change safety policies
transforms: [...] # vendor-specific adaptation
hardware_library: { ... } # chassis/linecard definitions

```

Three validation passes run at parse time:

1. **Schema validation:** field types, non-empty names, integer ranges. Unknown YAML keys are rejected at parse time, catching typos before execution.
2. **CEL selector compilation:** each selector string is compiled into an evaluable CEL program, surfacing syntax errors before execution.
3. **Layer ordering validation:** a forward scan ensures every requires entry names a previously-declared layer. This catches ordering bugs at parse time.

The optional imports field lists relative file paths to other blueprint files. At parse time, layers and assertions from imported files are merged into the importing blueprint, enabling reusable assertion libraries (e.g. a standard linting ruleset shared across projects).

Warning

Duplicate layer names are permitted. Multiple LayerSpec blocks with name: input are common in practice—each block adds primitives to the same layer. The requires mechanism enforces inter-layer ordering, not intra-layer uniqueness.

5.2 Selector DSL (CEL surface syntax)

Selectors identify which nodes or edges a primitive operates on. Conceptually, a selector is simply a boolean query over the current model: “for each node (or edge), should this primitive apply here?” The concrete surface syntax is a lightweight wrapper around Google’s Common Expression Language (CEL):

Listing 4. Selector syntax examples.

```

# Select all nodes
selector: "all()"

# Select nodes by property (e.g. region, role, type)
selector: "nodes[region=='eu-west']"
selector: "nodes[role=='core' && layer=='input']"
selector: "nodes[type=='Router' and region=='us-east']"

# Select edges
selector: "edges[src>=0]"
selector: "edges[true]"

```

The selector parser:

1. Identifies whether the selector targets nodes or edges and extracts the bracketed predicate.
2. Normalises minor syntax differences (e.g. `and` vs `&&`) so the predicate is unambiguous.
3. Compiles the predicate into an evaluable program.

At evaluation time, the selector ranges over all nodes (or edges) in the topology and returns the subset for which the predicate is true. This means selectors are always evaluated against the *current* model state: earlier primitives can add fields that later selectors use.

Mechanically, the compiler binds each entity’s properties as variables (`id`, `type`, `layer`, plus all `data_json` fields, flattened and accessible via dot notation) and evaluates the compiled predicate.

Note

Why missing fields are not errors. Because the model is layered, not every node has every field: a physical node will have interface and addressing facts, while a protocol node will have protocol-specific facts. Selectors must therefore tolerate heterogeneous populations. Mechanically, undeclared variable references (e.g., querying a field that some nodes lack) silently evaluate to false rather than producing an error. This lets a single selector range over mixed layers while still precisely matching only the nodes that carry the relevant facts.

For IP-valued fields, the selector also binds a `{field}_len` variable containing the prefix length, enabling selectors like `nodes[subnet_len == 30]`.

Note

Layer-aware selector patterns. Because the model is layered, robust selectors typically constrain both *what* a node is (role/type) and *where* it lives (layer). Common patterns include:

```
# Physical devices only
selector: "nodes[layer=='physical' && role=='core']"

# Protocol overlay nodes only (e.g. BGP layer)
selector: "nodes[layer=='bgp' && role=='core']"

# Edges within a layer (physical links, protocol sessions)
selector: "edges[layer=='physical']"

# Guard against heterogeneous fields (match only nodes that have the facts)
selector: "nodes[layer=='bgp' && has(node.bgp) && node.bgp.enabled==true]"
```

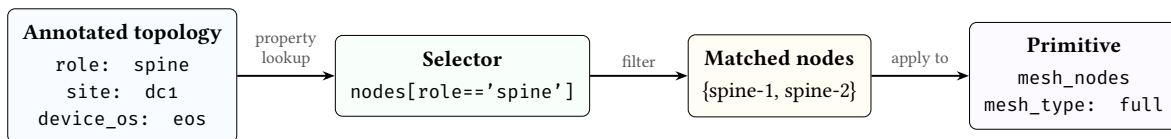


Figure 12. Selector-to-annotation data flow. Topology annotations are the values that selectors query; the selector filters nodes into a matched set; the primitive then operates on that set. The topology annotations are the “input language” and the selector is the bridge between topology intent and blueprint strategy.

5.3 Named groups

The `groups:` section defines reusable named node or edge sets. Group names are prefixed with `$` when referenced in selectors. Two forms are supported:

Listing 5. Group definitions: shorthand and full form.

```
groups:
  # Shorthand: just a selector string (e.g. by role)
  core_routers: "nodes[role=='core']"
  access_switches: "nodes[role=='access']"

  # Full form with kind and kind-specific fields
  dmz:
    selector: "nodes[zone=='dmz']"
    kind: security_zone
```

```
trust_level: 10
```

Groups are resolved before primitives execute. The selector parser expands `$group_name` references inline, and cycle detection rejects circular group dependencies at parse time. When blueprints import other files, groups from imported files are merged into the importing blueprint's namespace (name collisions are last-writer-wins).

The union operator (`|`) combines groups:

```
all_devices: $core_routers | $access_switches
```

This creates a set containing all nodes matching either group.

When a group has `kind: security_zone`, an optional `interface_selector` field may be supplied to express interface-level zone membership. This selector targets edges rather than nodes, enabling fine-grained policies where different interfaces on the same device belong to different zones:

```
dmz:
  selector: "nodes[zone=='dmz']"
  kind: security_zone
  trust_level: 10
  interface_selector: "edges[intf_role=='dmz']"
```

5.4 Named objects

The `objects:` section defines reusable address and service objects. These are referenced in zone policy and NAT policy rules using the `$` prefix (e.g. `$rfc1918`, `$web_services`). Both sub-maps default to empty, so omitting `objects:` is backward-compatible.

Listing 6. Named address and service objects.

```
objects:
  addresses:
    rfc1918:
      prefixes:
        - "10.0.0.0/8"
        - "172.16.0.0/12"
        - "192.168.0.0/16"
    corp_v6:
      prefixes_v6:
        - "2001:db8::/32"
  services:
    web_services:
      entries:
        - { protocol: tcp, port: 80 }
        - { protocol: tcp, port: 443 }
    dns:
      entries:
        - { protocol: udp, port: 53 }
        - { protocol: tcp, port: 53 }
```

`AddressObject` supports dual-stack addressing via separate `prefixes` (IPv4) and `prefixes_v6` (IPv6) fields. Prefixes are stored as strings and validated to `ipnet::IpNet` at object registry build time.

`ServiceObject` contains one or more `ServiceEntry` pairs of `protocol` and `port`. In zone policy or NAT rules, use `service: $web_services` in the `match:` block to reference a named service object.

5.5 Primitive catalogue

`NETCFG` provides twenty-three primitive types organised into five tiers, each distinguished by what it *mutates* in the network model.²

Note: primitive naming

The primitive types documented below use their stable DSL aliases (e.g. `allocate_resources`, `build_routing_policy`). The compiler internally uses canonical names (e.g. `provision_resources`, `policy_routing`); the DSL aliases are accepted for backward compatibility. Both names are valid in blueprint YAML; the canonical name appears in error messages and diagnostic output. Table 5 lists all renames.

²The `wasm_plugin` primitive requires the optional `wasm` Cargo feature; a default build exposes twenty-two primitives.

Table 5. Primitive canonical names and DSL aliases.

Canonical (Rust)	DSL alias	Tier
provision_resources	allocate_resources	Allocation
define_ip_pool	global_pool	Allocation
define_resource_pool	global_resource_pool	Allocation
policy_routing	build_routing_policy	Policy
policy_access	build_access_policy	Policy
policy_prefix_list	build_prefix_list	Policy
policy_community_list	build_community_list	Policy
policy_bgp_safety	generate_safe_bgp_filters	Policy
overlay_vxlan	build_vxlan	Overlay
overlay_evpn	build_evpn	Overlay

The five tiers are:

- **Topology** — create new nodes and edges (graph structure).
- **Allocation** — assign derived identifiers to existing nodes (IP addresses, ASNs, VLANs).
- **Policy** — compile high-level declarations into per-device rule sets (zone policies, ACLs, route-maps).
- **Overlay** — attach protocol-specific configuration stanzas (VXLAN, EVPN, BGP peering).
- **Meta** — annotate, gate, or extend the pipeline without producing device configuration.

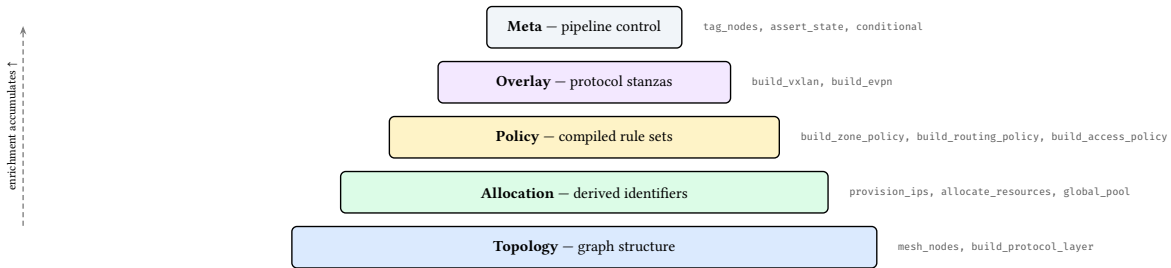


Figure 13. Primitive tier hierarchy. Lower tiers create structure; upper tiers annotate and configure existing nodes. Each tier builds on state written by earlier tiers.

All primitives follow the *select* → *enrich* → *validate* pattern: select a target population via graph queries, compute the enrichment, then write results to DeviceContext or the graph structure.

Table 6 summarises the catalogue. Full field tables follow.

Table 6. Primitive catalogue overview. Status: **Stable**, **Beta**, **Experimental**.

Primitive	Tier	Purpose	Status
mesh_nodes	Topology	Create edges between selected nodes (full, hub-and-spoke)	S
build_protocol_layer	Topology	Clone nodes into a named protocol overlay layer	S
provision_ips	Allocation	Allocate IP subnets to edges from a CIDR pool	S
allocate_resources	Allocation	Generic pool allocation for ASNs, VLANs, etc.	S
global_pool	Allocation	Register a named IP pool for hierarchical carving	S
global_resource_pool	Allocation	Register a named resource pool (non-IP)	S
build_routing_policy	Policy	Attach routing policy stanzas (route-maps)	S
build_access_policy	Policy	Attach access-control policy stanzas (ACLs)	S
build_prefix_list	Policy	Create named prefix-list entries on matched nodes	S
build_community_list	Policy	Create named community-list entries on matched nodes	S
generate_safe_bgp_filters	Policy	Auto-generate RFC 1918 bogon filters	S
build_zone_policy	Policy	Zone-based firewall policies with ordered rules	B
build_nat_policy	Policy	NAT policies (SNAT interface/pool, DNAT port-forward)	B
build_qos_policy	Policy	QoS policies (classifier, scheduler, policer, shaper)	B
build_vxlan	Overlay	Assign VXLAN VNI and multicast group bindings	S
build_evpn	Overlay	Assign EVPN route-distinguisher and route-target	S
tag_nodes	Meta	Stamp key-value metadata onto matched nodes	S
map_hardware_inventory	Meta	Assign chassis model and line-card slot mappings	S
assert_state	Meta	Mid-pipeline assertion checkpoint	S
conditional	Meta	CEL-based branching within blueprints	S
custom_primitive	Meta	Named reusable primitive group with parameters	S
inject_secrets	Meta	Read environment variables into node metadata	S
wasm_plugin	Meta	Execute a WebAssembly plugin against matched nodes	E

5.5.1 mesh_nodes

Creates edges between selected nodes. Supports two mesh topologies: `full` ($\binom{n}{2}$ edges) and `hub_and_spoke` (hubs full-meshed, spokes connected to all hubs). Idempotent: existing edges are detected via a `HashSet<(i32, i32)>` and skipped.

Table 7. mesh_nodes fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for nodes to mesh
mesh_type	String	Yes	<code>full</code> or <code>hub_and_spoke</code>
hub_selector	String	H&S	Selector for hub nodes (hub-and-spoke only)
spoke_selector	String	H&S	Selector for spoke nodes (hub-and-spoke only)

Example: full mesh within a site

```
- type: mesh_nodes
  selector: "nodes[site='a']"
  mesh_type: full
```

With 4 nodes at site A, this creates $\binom{4}{2} = 6$ edges and assigns interface names `eth0`, `eth1`, etc. to each node's `peer_interfaces` map.

5.5.2 provision_ips

Allocates subnets from a CIDR pool to edges. See §7.1 for the full five-pass algorithm.

Table 8. provision_ips fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for edges
pool	String	Yes	CIDR (e.g. 10.0.0.0/24) or global pool name
subnet_size	u8	No	Prefix length per edge (default: 30 for IPv4, 126 for IPv6)
ipv6_pool	String	No	IPv6 CIDR or global pool name for dual-stack
ipv6_subnet_size	u8	No	IPv6 prefix length per edge
strategy	String	No	dense (default), sparse, or hash
pool_size	u8	No	When referencing a global pool, prefix length of block to carve
vrf	String	No	VRF domain (default: default)

5.5.3 build_protocol_layer

Clones physical nodes into a named protocol layer. See §7.2 for the overlay construction mechanism.

Table 9. build_protocol_layer fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for source-layer nodes
layer	String	Yes	Target layer name (must differ from source)
config	JSON	Yes	Config overlay deep-merged into cloned nodes
clone_underlying	bool	No	Also clone edges between selected nodes (default: false)

5.5.4 allocate_resources

Generic pool allocator for non-IP resources (ASNs, VLANs, loopback IDs).

Table 10. allocate_resources fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for nodes
resource_type	String	Yes	Key name in DeviceContext.metadata
pool	String	Yes	Range (e.g. 65000-65999) or global pool name
strategy	String	No	dense (default), sparse, or hash
pool_size	u32	No	When referencing a global pool, number of values to carve

5.5.5 global_pool

Registers a named IP address pool at blueprint scope, enabling hierarchical pool carving across multiple provision_ips primitives.

Table 11. global_pool fields.

Field	Type	Required	Description
name	String	Yes	Pool identifier referenced by provision_ips
pool	String	Yes	CIDR range (e.g. 10.0.0.0/8)
vrf	String	No	VRF domain (default: default)

5.5.6 global_resource_pool

Registers a named non-IP resource pool at blueprint scope.

Table 12. global_resource_pool fields.

Field	Type	Required	Description
name	String	Yes	Pool identifier
resource_type	String	Yes	Resource category (e.g. asn)
pool	String	Yes	Value range (e.g. 65000-65999)

5.5.7 conditional

CEL-based branching. The condition is evaluated as a CEL expression; if true, then_primitives execute, otherwise else_primitives (if present) execute.

Table 13. conditional fields.

Field	Type	Required	Description
condition	String	Yes	CEL expression
then_primitives	Vec<Primitive>	Yes	Primitives if condition is true
else_primitives	Vec<Primitive>	No	Primitives if condition is false

5.5.8 custom_primitive

A named, reusable group of primitives with parameters. The parameters map is pushed onto a stack and accessible within the nested primitives.

Table 14. custom_primitive fields.

Field	Type	Required	Description
name	String	Yes	Primitive name (for reporting)
parameters	map (string → value)	Yes	Named parameters
primitives	list of primitives	Yes	Nested primitive list

5.5.9 inject_secrets

Reads environment variables into node metadata. Keys in the secrets map are metadata field names; values are environment variable names.

Table 15. inject_secrets fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
secrets	map (string → string)	Yes	{metadata_key: env_var_name}

5.5.10 build_routing_policy

Attaches routing policy stanzas to matched nodes. Each statement defines a route-map entry with match conditions and set actions, stored as a Stanza in DeviceContext.stanzas for downstream template rendering.

Table 16. build_routing_policy fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
policy_name	String	Yes	Policy name (e.g. BGP-EXPORT)
statements	list of statements	Yes	Ordered route-map entries

Each RoutingPolicyStatement has fields: name (sequence label), action (permit or deny), and optional match/set clauses: match_prefix_list, match_community_list, match_as_path,

set_local_preference (integer), set_metric (integer), set_as_path_prepend, set_community, and next_hop (IP address).

Example: BGP export policy

```
- type: build_routing_policy
  selector: "nodes[role='border']"
  policy_name: "BGP-EXPORT"
  statements:
    - name: "10"
      action: permit
      match_prefix_list: "CUSTOMER"
      set_local_preference: 100
    - name: "20"
      action: deny
```

5.5.11 build_access_policy

Attaches access-control policy stanzas to matched nodes. Each rule defines a firewall or ACL entry with source, destination, protocol, and port fields.

Table 17. build_access_policy fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
policy_name	String	Yes	Policy name (e.g. EDGE-ACL)
rules	list of rules	Yes	Ordered ACL entries

Each AccessPolicyRule has fields: name, action (permit/deny), optional source_prefix (CIDR), destination_prefix (CIDR), protocol (string), source_port (integer), and destination_port (integer). Both policy primitives enforce node targeting—policies cannot target edges.

Example: DMZ access control list

```
- type: build_access_policy
  selector: "nodes[zone='dmz']"
  policy_name: "UNTRUSTED-TO-INTERNAL"
  rules:
    - name: "100"
      action: deny
      protocol: tcp
      destination: "10.0.0.0/8"
    - name: "999"
      action: permit
```

5.5.12 wasm_plugin

[Experimental] Executes a WebAssembly plugin binary against matched nodes. Requires the wasm Cargo feature flag (`-features wasm`); without it, the pipeline returns a descriptive error. Uses the wasmtime [14] v18 runtime.

Table 18. wasm_plugin fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
plugin_path	String	Yes	Filesystem path to compiled .wasm binary

The execution pipeline: (1) load and compile the .wasm module, (2) instantiate per matched node without host imports, (3) resolve the apply_node exported symbol, (4) pass the node's DeviceContext as serialised JSON. The current implementation validates compilation and symbol resolution; full linear memory I/O for JSON exchange is planned for a future release (see §15).

Note

The wasm feature is optional to avoid the wasmtime dependency (which adds ~40 MB to the binary) for users who do not need plugin extensibility.

5.5.13 build_prefix_list

Creates named prefix-list entries on matched nodes, stored as Stanza objects for downstream template rendering.

Table 19. build_prefix_list fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
prefix_list_name	String	Yes	Name of the prefix list
entries	Vec<PrefixListEntry>	Yes	Ordered prefix-list entries

Each PrefixListEntry has: prefix (CIDR string), action (permit/deny), optional le (less-than-or-equal prefix length), and optional ge (greater-than-or-equal prefix length).

5.5.14 build_community_list

Creates named community-list entries on matched nodes.

Table 20. build_community_list fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
community_list_name	String	Yes	Name of the community list
entries	Vec<CommunityListEntry>	Yes	Ordered community-list entries

Each CommunityListEntry has: community (e.g. 65001:1000) and action (permit/deny).

5.5.15 generate_safe_bgp_filters

Auto-generates RFC 1918 bogon filters as prefix-list and routing-policy stanzas on matched nodes.

Table 21. generate_safe_bgp_filters fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
prefix_list_name	String	No	Name for the generated prefix list
policy_name	String	No	Name for the generated routing policy
block_rfc1918	bool	No	Block RFC 1918 prefixes (default: true)
permit_default	bool	No	Add a trailing permit-any (default: false)

5.5.16 build_zone_policy

Compiles a zone-based firewall policy for traffic flowing between two security zones. The operator declares the zone pair, an ordered list of match/action rules, and a default action (deny by default). At execution time, the primitive validates zone names against the group registry, constructs a typed SecurityModel on each matched node, and appends the zone policy to the node's DeviceContext. Vendor templates render the policy into platform-specific syntax (Junos security-zone ACLs, Cisco ZBFW class-maps, VyOS zone-policy rules, etc.).

Table 22. build_zone_policy fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target devices
from_zone	String	Yes	Source security zone name
to_zone	String	Yes	Destination security zone name
default_action	String	No	permit or deny (default: deny)
rules	Vec<Rule>	Yes	Ordered list of match/action rules

Each rule has a name, an action (permit/deny), and an optional match block with fields: source, destination, service, protocol, destination_port, icmp_type. Optional log and stateful flags control logging and connection-tracking behaviour.

5.5.17 build_nat_policy

Attaches NAT policy rules to matched devices. Supports four rule types: snat (source NAT via interface or pool), dnat (destination NAT / port-forwarding), nat64 (RFC 6146 stateful IPv6-to-IPv4 translation), and nptv6 (RFC 6296 IPv6 prefix translation). Rules are ordered and may optionally reference zone pairs (from_zone, to_zone) for zone-scoped NAT.

Table 23. build_nat_policy fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target devices
rules	Vec<NatRule>	Yes	Ordered list of NAT rules

Each NatRule has a name, a type (snat/dnat/nat64/nptv6), an optional match block, and a type-specific parameter block. SNAT rules specify mode (interface or pool) and optional pool_addresses. DNAT rules specify external_address, optional external_port, internal_address, and optional internal_port. NAT64 rules specify a nat64_prefix; NPTv6 rules specify internal_prefix, external_prefix, and an optional allow_best_effort flag. Rules may also carry from_zone/to_zone for zone-scoped NAT and an optional log flag.

The match block accepts four optional fields:

Table 24. NAT rule match fields (NatMatchSpec).

Field	Type	Description
source	String	Source CIDR or \$address_object_name
destination	String	Destination CIDR or \$address_object_name
service	String	\$service_object_name or bare identifier
protocol	String	IP protocol (e.g. tcp, udp)

5.5.18 build_qos_policy

Attaches Quality of Service policies to matched nodes. Each policy defines a set of traffic classes (classifier, marker, scheduler, policer, shaper) and a set of interface bindings that apply policies in the ingress or egress direction.

Table 25. build_qos_policy fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
policies	map (name → QosPolicy)	No	Named QoS policies
interfaces	map (name → binding)	No	Per-interface policy bindings

A QosPolicy contains a map of named TrafficClass entries. Each traffic class may specify a classifier (matching on DSCP, CoS, protocol, addresses, or ports), a marker (rewriting DSCP/CoS), a scheduler (priority queuing, weighted fair queuing, bandwidth guarantees), a policer (CIR/burst enforcement), and a

shaper (output rate limiting). An InterfaceBinding specifies an input_policy and/or output_policy name.

Example: data-centre QoS policy

```
- type: build_qos_policy
  selector: "nodes[role=='leaf']"
  policies:
    DC-QoS:
      classes:
        voice:
          classifier: { dscp: "ef" }
          scheduler: { priority: true, bandwidth_percent: 10 }
          policer: { cir: 1000000, exceed_action: drop }
        bulk:
          classifier: { dscp: "af11" }
          scheduler: { weight: 20 }
          shaper: { shape_rate: 500000000 }
      interfaces:
        Ethernet1:
          input_policy: DC-QoS
          output_policy: DC-QoS
```

Note

The device_os annotation determines platform-specific QoS semantics. For example, NX-OS devices enforce a maximum of 8 traffic classes per policy and require explicit system-qos activation.

5.5.19 build_vxlan

Assigns VXLAN tunnel bindings to matched nodes. VNIs are allocated sequentially from vni_base.

Table 26. build_vxlan fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for VTEP nodes
vni_base	u32	Yes	Starting VNI (e.g. 10000)
mcast_group_base	String	No	Multicast group for BUM replication

5.5.20 build_evpn

Assigns EVPN route-distinguisher (RD) and route-target (RT) values to matched nodes.

Table 27. build_evpn fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for EVPN nodes
route_distinguisher_base	String	Yes	RD base (e.g. auto)
route_target_base	String	Yes	RT base (e.g. 65001:10000)

5.5.21 tag_nodes

Stamps flat key-value metadata onto matched nodes. Tags are merged into each node's data_json and are available to subsequent selectors and assertions.

Table 28. tag_nodes fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
tags	map (string → value)	Yes	Key-value pairs to stamp

5.5.22 map_hardware_inventory

Assigns chassis model and line-card slot mappings to matched nodes. Used by the TSDM layer (§7.4) via CEL extension functions (e.g., `hw.port_name(node.chassis, slot)`) to resolve abstract interface names into physical slot/port identifiers.

Table 29. map_hardware_inventory fields.

Field	Type	Required	Description
selector	String	Yes	CEL selector for target nodes
chassis_model	String	Yes	Chassis identifier (e.g. dcs-7508)
slots	map (integer → string)	Yes	Slot → line-card model mapping

5.5.23 assert_state

A mid-pipeline assertion checkpoint. Syntactically identical to a top-level `AssertionSpec` but placed inline within a layer's primitive list. This allows the operator to validate intermediate state between primitives rather than waiting until the post-execution assertion pass.

Table 30. assert_state fields.

Field	Type	Required	Description
name	String	Yes	Assertion name (for diagnostics)
severity	String	No	error (default) or warning
select	String	Yes	CEL selector for items to check
check	AssertionCheck	Yes	Predicate (same types as top-level assertions)
help	String	No	Remediation hint shown on failure

5.6 Design-rule assertions

Blueprints may declare assertions: post-execution invariants checked after all primitives complete but before commit. Each assertion has a severity: error (pipeline failure) or warning (diagnostic only).

Listing 7. Design-rule assertions in a blueprint.

```
assertions:
- name: "all routers have hostname"
  severity: error
  select: all()
  check: { type: field_exists, field: hostname }
- name: "IPs within allocation"
  severity: error
  select: "nodes[src_ip != null]"
  check:
    field_in_cidr:
      field: src_ip
      cidr: "10.0.0.0/8"
- name: "unique hostnames per site"
  severity: warning
  select: all()
  check:
    unique_per_group:
      field: hostname
      group_by: site
- name: "private ASN range"
  severity: warning
  select: "nodes[asn != null]"
  check:
    custom_cel:
      expression: "asn >= 64512 && asn <= 65534"
```

Nineteen check types are available:

Table 31. Assertion check types.

Check	Semantics
field_exists	Every matched node has the named field in its data_json
min_count	The matched set contains at least N items
max_count	The matched set contains at most N items
min_edges	Every matched node has at least N outgoing edges
unique_per_group	Within groups defined by group_by, the field value is unique
field_in_cidr	Every matched node's IP field is contained within the specified CIDR
custom_cel	Arbitrary CEL expression evaluated per matched entity; fails if false
use_rule	Reference a named CEL macro from the rules: section
is_connected	The matched subgraph (nodes + interconnecting edges) is connected
is_acyclic	The matched subgraph contains no cycles
is_bipartite	The matched subgraph is bipartite (two-colourable)
reachability	Every matched node can reach at least one node in a target selector
match_schema	Each matched node's data_json validates against a JSON Schema
zone_membership	Every matched node belongs to one of the declared security zones
zone_policy_exists	A zone policy exists for the specified from/to zone pair
no_shadowed_rules	No rule in a zone policy is fully shadowed by a preceding rule
zone_pairs_covered	All declared zone pairs have corresponding zone policies
zone_policy_permits	A specific traffic 5-tuple is permitted by the zone policy
zone_policy_denies	A specific traffic 5-tuple is denied by the zone policy

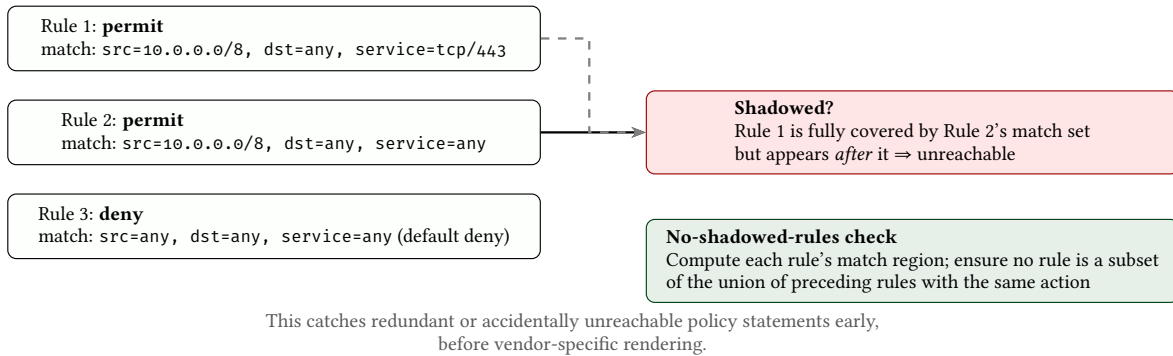


Figure 14. Visual intuition for no_shadowed_rules. The assertion flags rules whose match space is completely covered by earlier rules, making them unreachable (and often indicating operator intent errors).

This enables “policy as code” directly in the blueprint. Assertions are evaluated cross-layer (no layer filter), so a single assertion can validate properties across both physical and protocol layers.

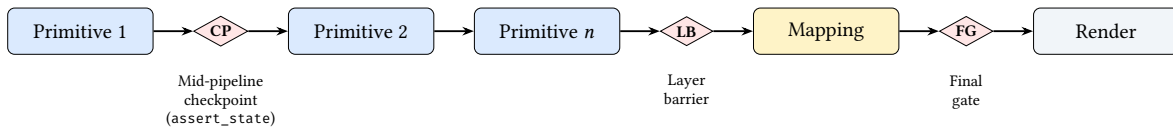


Figure 15. Assertion firing points. **CP**: mid-pipeline checkpoint (inline assert_state); **LB**: layer barrier (runs after a layer's primitives complete); **FG**: final gate (runs before rendering). All three evaluate the same assertion types but fire at different pipeline positions for earlier feedback.

5.7 Named rules (CEL macros)

The rules: section defines named CEL macros that can be referenced by assertions via the use_rule check type. This avoids duplicating complex CEL expressions across multiple assertions.

Listing 8. Named rules and their use in assertions.

```
rules:
```



```

tenant-assigned: "has(node.tenant) && node.tenant != ''"
vrf-configured: "has(node.vrf_name) && node.vrf_name != ''"
leaf-has-vlan: "has(node.vlan_id) && node.vlan_id > 0 && node.vlan_id < 4095"

```

Rules are simple string-valued CEL expressions stored as a name-to-expression map. At evaluation time, a `use_rule` check looks up the named rule and evaluates it as though it were an inline `custom_cel` expression. Rules from imported files are merged into the importing blueprint's rule namespace.

5.8 Blast radius policies

The `blast_radius:` section declares change-safety policies that constrain how much the topology may change in a single commit. Each policy specifies a check against the computed `MutationPlan`:

Table 32. Blast radius check types.

Check	Semantics
<code>max_deletions</code>	At most N nodes or edges may be deleted
<code>max_modifications</code>	At most N nodes or edges may be modified
<code>max_percentage_changed</code>	At most $P\%$ of the topology may be altered (0.0–100.0)

Table 33. `BlastRadiusSpec` fields.

Field	Type	Required	Description
<code>name</code>	String	Yes	Policy name (for reporting)
<code>select</code>	String	No	CEL selector to scope the policy; omit for global
<code>check</code>	object	Yes	One of the check types above, with a <code>value</code> field

When `select` is omitted, the policy applies to the entire mutation plan.

Listing 9. Blast radius policies.

```

blast_radius:
- name: tenant-vrf-protection
  check:
    type: max_deletions
    count: 3
- name: fabric-change-cap
  check:
    type: max_percentage_changed
    value: 25.0

```

5.9 Target-specific transforms

The `transforms:` section declares AST-to-AST rewrite rules that adapt vendor-neutral stanzas into platform-specific syntax (§7.4 details the execution engine). Each `TransformationSpec` has a `name`, an optional `when` predicate (CEL expression evaluated per device), and a sequence of `RewriteRule` entries:

Table 34. Transform rewrite rule fields.

Field	Type	Required	Description
<code>name</code>	String	Yes	Transform name
<code>when</code>	String	No	CEL predicate (e.g. <code>device_os == 'nxos'</code>)
<code>rules</code>	<code>Vec<RewriteRule></code>	Yes	Sequence of match/apply rewrites

Each `RewriteRule` has a `match_expr` (CEL expression selecting stanzas) and an apply map (field name → CEL expression computing the new value):

Listing 10. Vendor-specific transform.

```

transforms:
- name: nxos-interface-rewrite
  when: "device_os == 'nxos'"

```

```

rules:
  - match_expr: "stanza.kind == 'interface'"
    apply:
      name: "'Ethernet1/' + string(stanza.fields.abstract_index + 1)"

```

6 Compilation Pipeline

NETCFG’s compilation pipeline has five distinct layers of abstraction, operating as a true network compiler. Figure 16 shows the data flow from abstract intent to concrete syntax.

A key property of this pipeline is **determinism**: configuration is a pure function $f(\text{Annotated Topology, Blueprint}) = \text{DeviceIR} \rightarrow \text{Config}$. If neither the topology annotations nor the blueprint strategies have changed, regenerating the configuration produces a bit-identical result, making any divergence immediately detectable via diff. Combined with the structural diff engine (§7.3), this enables a “regenerate and compare” workflow for detecting configuration drift without runtime monitoring.

Note

The conference paper describes six implementation stages (parse, shadow, execute, map, render, write); this report groups these into five conceptual layers (Intent, Assertions, Mapping/DeviceIR, TSDM, Syntax). Both describe the same pipeline at different levels of abstraction: the conference paper’s “parse” and “shadow” stages are subsumed by Layer 1 (Intent Transformation), and “render” and “write” are subsumed by Layer 5 (Syntax Serialisation).

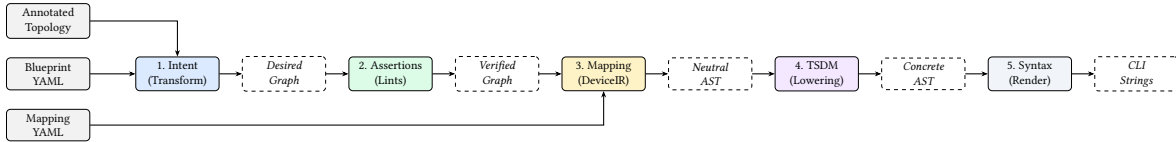


Figure 16. NetCfg compilation pipeline. High-level intent is progressively lowered through three distinct AST representations (Verified Graph, Neutral AST, Concrete AST) before serialisation.

6.1 Layer 1: Intent Transformation

The blueprint is parsed into an AST and its primitives are applied to a transactional *shadow* copy of the annotated topology (Figure 17). The shadow is created via a serialisation round-trip, providing an isolated working copy: if any primitive or assertion fails, the shadow is discarded and the production graph remains untouched. Each primitive’s selectors reference annotations already present on the topology nodes (e.g. role, site, device_os); the primitive then derives new state from those annotations and writes it back to the shadow.

Composable primitives execute in declaration order within each layer. Topology primitives (mesh_nodes, build_protocol_layer) create graph structure; allocation primitives (provision_ips, allocate_resources) assign derived identifiers; policy primitives (build_zone_policy, build_routing_policy) compile intent into per-device rule objects. Each primitive follows the *select* → *enrich* → *validate* pattern (§2.7), and each sees the enriched model left by its predecessors.

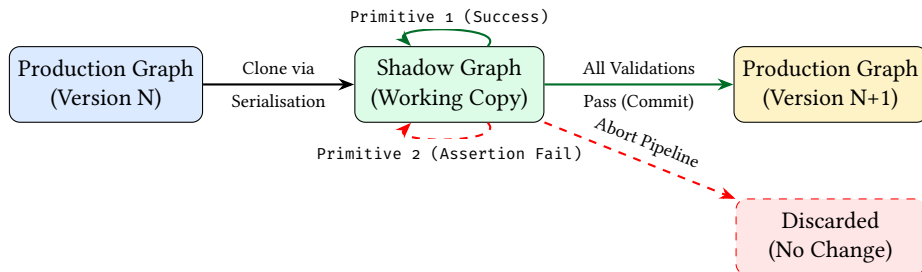


Figure 17. Transactional Safety via the Shadow Topology. Mutations occur on an isolated clone. If any primitive or assertion fails, the shadow is discarded without side-effects on the production graph.

6.2 Layer 2: Topological Assertions

After all primitives complete, the DSL-driven assertion engine (§5.6) evaluates every assertion declared in the blueprint against the enriched graph. Assertions are typed checks—field existence, count bounds, CIDR

containment, graph connectivity, zone-pair coverage, and arbitrary CEL predicates—evaluated cross-layer so that a single assertion can span physical and protocol layers simultaneously.

Assertions with severity error abort the pipeline; those with severity warning emit diagnostics but allow compilation to continue. This layer acts as the quality gate between intent transformation and lowering; only a fully validated model proceeds to DeviceIR generation.

6.3 Layer 3: DeviceIR and the Mapping DSL

Between stage 2 (Assertions) and stage 4 (TSDM) sits the Mapping Evaluation engine. A Mapping DSL lowers the rich topology graph into a vendor-neutral **Abstract Syntax Tree (DeviceIR)**. This representation captures logical intent (e.g. "Neighbor X is remote-as Y") without hardware context.

A mapping rule consists of a selector and a stanza template. When evaluated against the graph, it emits discrete configuration units (e.g., interfaces, BGP neighbours) that are attached to each node's DeviceContext.stanzas list (Figure 18).

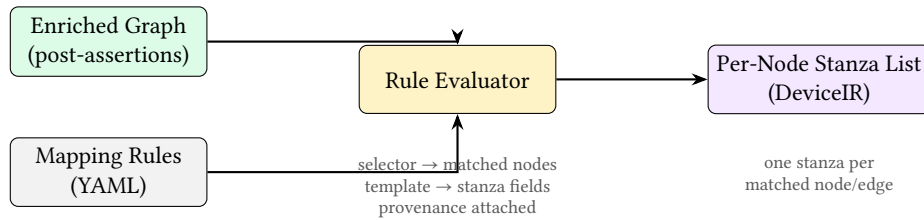


Figure 18. Mapping evaluation. Each rule's selector matches a subset of the enriched graph; the stanza template extracts fields into a structured DeviceIR unit attached to the node's DeviceContext.

Listing 11. Mapping DSL extracting an interface stanza.

```

rules:
- selector: "edges[layer=='physical']"
  stanza:
    kind: "interface"
    fields:
      name: "{{ node.peer_interfaces[peer.hostname] }}"
      description: "Link to {{ peer.hostname }}"
      mtu: 9000
  
```

6.4 Layer 4: Target-Specific Transform (TSDM)

The **TSDM Layer** performs AST-to-AST transformations (Figure 19). It lowers the Neutral AST into a platform-specific AST, remapping logical interfaces to physical ports (e.g. Eth1 → Eth1/1/1) based on the target's chassis model and hardware layout.

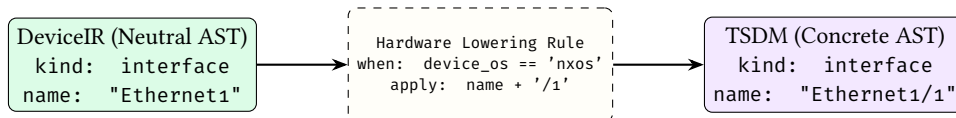


Figure 19. AST-to-AST lowering via TSDM. Vendor-neutral DeviceIR is transformed into a target-specific data model using rules evaluated against the device's metadata.

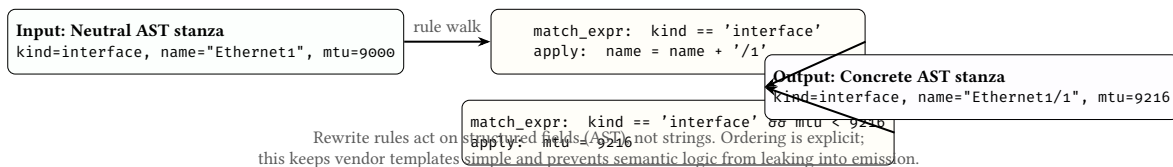


Figure 20. TSDM as a rewrite system. Rules match stanzas structurally and apply deterministic field mutations, producing a concrete AST suitable for vendor-specific rendering without embedding hardware logic in templates.

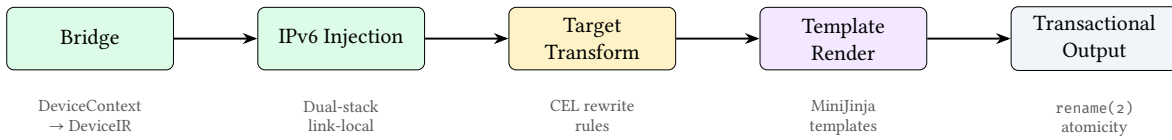


Figure 21. TSDM lowering pipeline. DeviceIR stanzas pass through bridge translation, optional IPv6 injection, target-specific AST rewriting, template rendering, and transactional file output.

6.5 Layer 5: Syntax Serialisation

MiniJinja templates act as a *dumb emission layer*, serialising the final Target-Specific AST into vendor-specific CLI syntax. Templates contain no conditional logic beyond iteration over stanza lists and field interpolation—all architectural decisions have already been resolved by the preceding layers. This keeps templates thin and auditable: a template bug can only affect formatting, never topology or policy semantics.

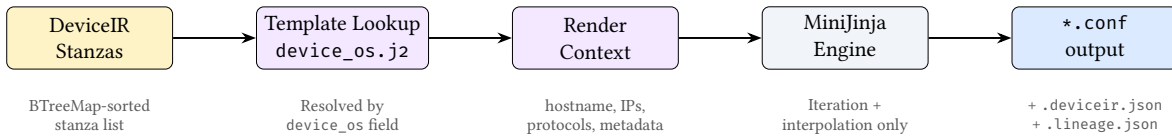


Figure 22. Template rendering pipeline. DeviceIR stanzas are resolved to a vendor template, combined with a render context, and emitted by the MiniJinja engine as configuration text plus audit artefacts.

The TransactionalOutput writer collects all files in temporary locations (using `NamedTempFile` in the same directory for same-filesystem guarantee), then atomically renames all via POSIX `rename(2)` on commit. If any stage fails, all temp files are dropped without committing—no partial output directory is ever visible.

Three artifacts are emitted per node:

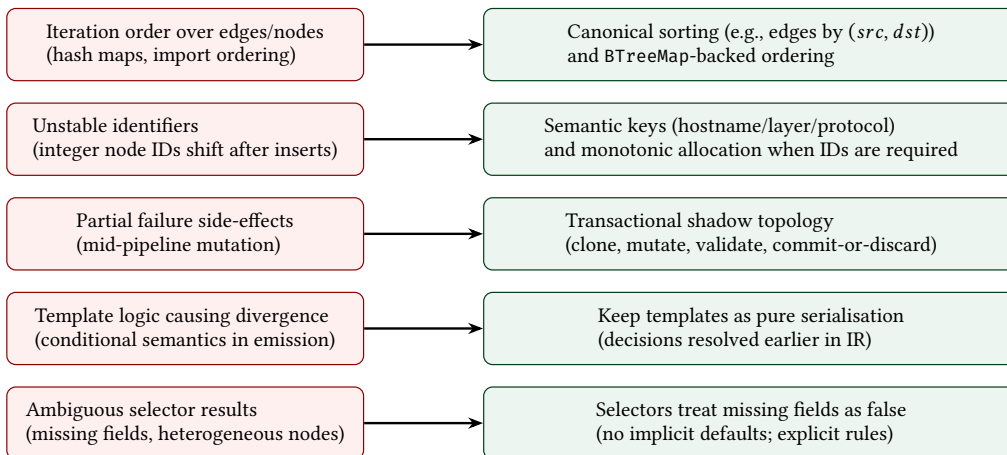
- `{hostname}.conf`: the rendered device configuration.
- `{hostname}.deviceir.json`: the raw DeviceIR stanzas (audit).
- `{hostname}.lineage.json`: per-line mapping from config lines to the `rule_id` that produced them (provenance tracing).

6.5.1 Determinism guarantee

Given a fixed annotated topology G , blueprint B , and mapping M , the output configuration set $C = \text{compile}(G, B, M)$ is deterministic. Serialisation round-trip produces a bit-identical shadow. Primitives execute in declaration order. Selectors evaluate deterministically. Node ID allocation is monotonic. IP allocation iterates edges sorted by (src, dst) . DeviceIR stanzas use `BTreeMap` and are post-sorted. Template rendering is pure.

Potential nondeterminism

Stabilisation mechanism



Combined, these constraints make compilation a pure function of inputs: $\text{same } G, B, M \Rightarrow \text{bit-identical outputs}$.

Figure 23. Determinism as a set of enforced constraints. Each common source of nondeterminism is eliminated by canonical ordering, semantic keys, transactional isolation, and keeping the render layer free of semantic logic.

7 Core Mechanisms

7.1 Five-pass IPAM (provision_ips)

The IP allocator implements a five-pass algorithm:

1. **State discovery:** Scan all nodes for existing IP assignments (`src_ip`, `dst_ip`, `subnet` fields).
2. **Edge selection:** Evaluate the CEL selector; sort matched edges by (`src`, `dst`) for determinism.
3. **Pre-reservation:** Insert existing allocations into the allocator, preventing re-allocation and enforcing stickiness.
4. **Fulfilment:** For each edge, allocate a subnet using the configured strategy (dense, sparse, or hash). Extract host addresses from the subnet.
5. **Write-back:** Write `src_ip`, `dst_ip`, `subnet` (and IPv6 equivalents for dual-stack) into each endpoint's `DeviceContext`.

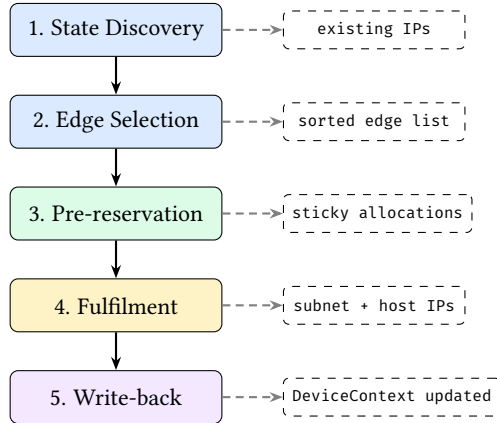
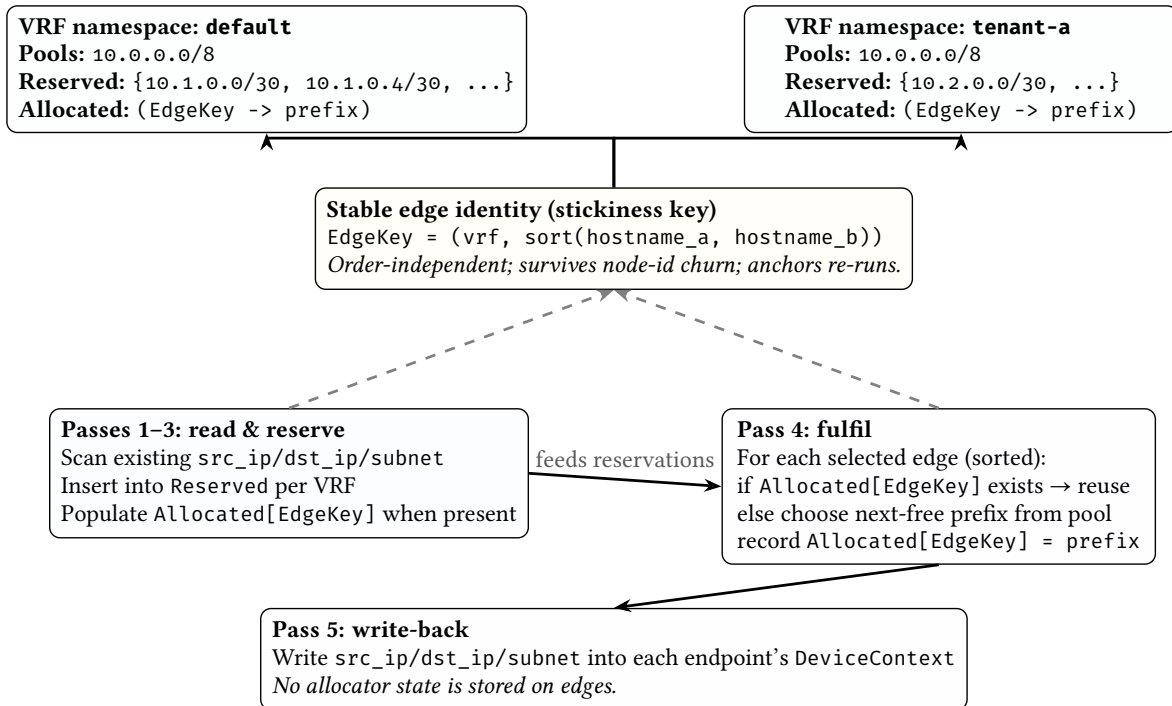


Figure 24. Five-pass IPAM algorithm. Passes 1–3 gather state and reserve existing allocations; pass 4 allocates new subnets; pass 5 writes results to `DeviceContext`. The two-phase structure (read then write) satisfies Rust's ownership rules without unsafe code.



Overlaps are allowed across VRFs; conflicts within a VRF are rejected.

Figure 25. Allocator state model. Allocations are partitioned by VRF and keyed by a stable semantic edge identity (hostnames, not integer IDs). Existing allocations are pre-reserved for stickiness; new prefixes are chosen from the free space within the VRF namespace, then written back to node `DeviceContext`.

Identity-based stickiness. The allocator uses a canonical identity —`sorted(hostname_a, hostname_b)`— as the primary key. If a prior allocation exists for the same identity, the same prefix is returned without pool search. This anchors allocations to stable semantic identity, not integer IDs. If topology IDs change (e.g., after a re-import) but hostnames remain the same, allocations are preserved.

VRF isolation. The allocator maintains per-VRF state. Two primitives using the same CIDR pool in different VRFs can allocate overlapping subnets without conflict. Pool overlap *within* the same VRF is detected and rejected.

Allocation strategies:

- **dense:** Sequential allocation from the pool start. First edge gets the first available subnet.
- **sparse:** Skips every other subnet, leaving growth headroom between allocations.
- **hash:** Subnet position is derived from a hash of the edge identity, producing topology-independent allocations that do not depend on evaluation order.

Hierarchical pools. A `global_pool` primitive registers a large CIDR (e.g., `10.0.0.0/8`). Subsequent `provision_ips` primitives reference the pool by name and specify a `pool_size` (e.g., 24), carving /24 blocks from the parent pool on demand.

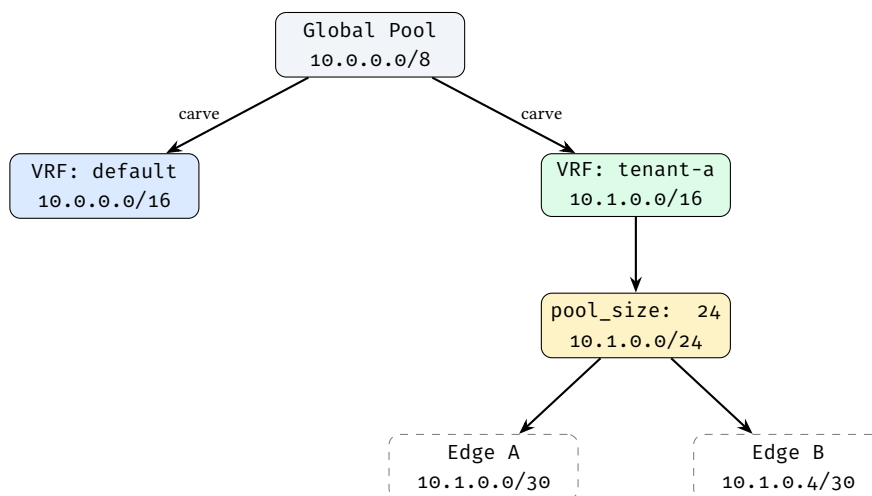


Figure 26. Hierarchical IP pool carving with VRF isolation.

Dual-stack. When `ipv6_pool` is specified, both IPv4 and IPv6 addresses are allocated in a single primitive invocation. IPv6 fields (`src_ipv6`, `dst_ipv6`, `subnet_v6`) are written alongside IPv4 fields.

7.2 Protocol overlay construction (`build_protocol_layer`)

`build_protocol_layer` clones physical nodes into a named protocol layer. For each selected source-layer node:

1. Allocate a new monotonic node ID.
2. Add the node to the target layer.
3. Deep-merge the config overlay into the cloned node's metadata.
4. Record `_source_id` as a parent mapping. For MLAG/vPC setups, `_source_id` can be an array mapping one logical protocol node to multiple physical host switches.
5. Optionally (`clone_underlying: true`) clone edges between source-layer nodes into the overlay.

Cross-layer traceability: the `_source_id` field links each protocol node to its physical origin. Since `provision_ips` already enriched the physical node's DeviceContext with IP addresses, the cloned protocol node inherits them via deep merge—this is how IP addresses flow from IPAM allocation to BGP configuration without explicit wiring.

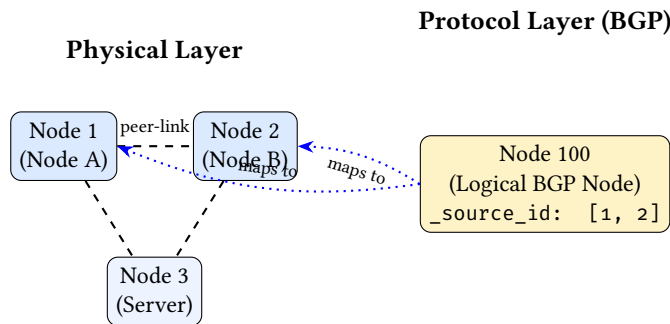
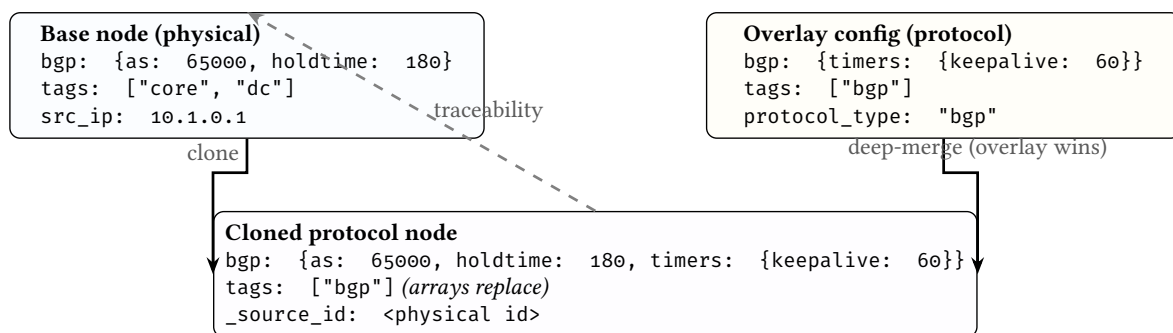


Figure 27. MLAG/vPC topology mapping. Two physical nodes collapse into a single logical protocol node via array `_source_id`.

Deep merge semantics: When both the base node and the config overlay contain nested JSON objects under the same key, the merge recurses key-by-key (overlay wins on conflict). Arrays are replaced entirely rather than concatenated, ensuring idempotency. This preserves sibling keys that the overlay does not mention. For example, a base node with `{bgp: {as: 65000, holdtime: 180}}` and an overlay with `{bgp: {timers: {keepalive: 60}}}` produces `{bgp: {as: 65000, holdtime: 180, timers: {keepalive: 60}}}`.

Overlay clone + deep-merge semantics



Objects merge recursively; scalar conflicts resolved by overlay; arrays replaced to preserve idempotency.

Figure 28. Protocol overlay construction as a clone + deep merge. The cloned node inherits prior enrichment (e.g., IPs) from the physical node and overlays protocol-specific configuration; the `_source_id` link provides cross-layer traceability without cross-layer edges.

Idempotency: The primitive scans the target layer for existing nodes with `_source_id` matching source nodes. Already-cloned nodes are skipped, so re-running the primitive after a partial failure produces the correct result.

7.3 Configuration diffing (DiffEngine)

The diff engine computes a *mutation plan* between two topologies by exporting each layer to JSON and comparing:

1. **Node diff:** Per layer, compare node sets by a composite semantic key (layer, hostname, protocol_type). Diffing by integer ID is avoided, as node insertions would shift downstream IDs and create massive false diffs. Detect additions, deletions, and modifications (field-by-field property comparison via `extract_properties`, which unpacks the `data_json` blob).
2. **Edge diff:** Compare by (`src`, `dst`) pair with structural property comparison.

The `MutationPlan` contains six change types: `AddNode`, `RemoveNode`, `UpdateNode`, `AddEdge`, `RemoveEdge`, `UpdateEdge`. Changes are sorted deterministically by (change_type, layer, id/src/dst) for reproducible output.

The `plan` command displays the plan as a human-readable diff; the `ci-run` command uses it to compute per-device configuration diffs.

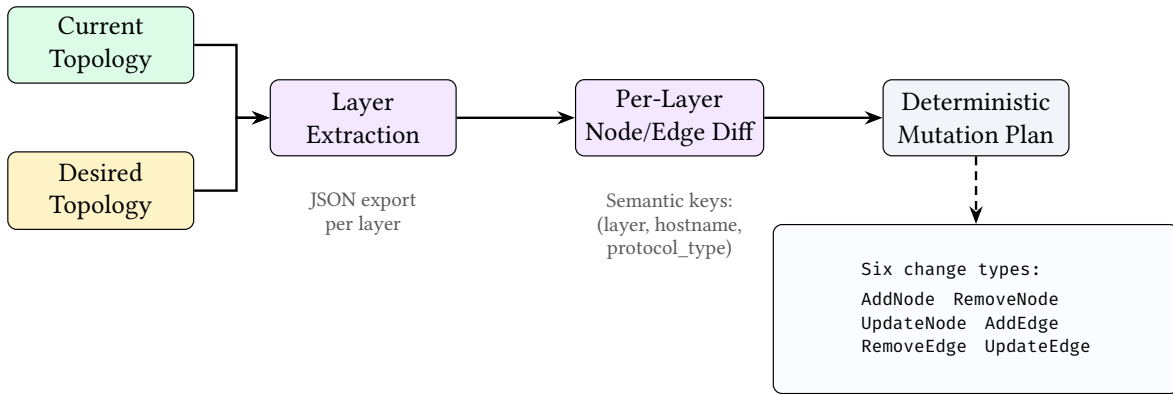
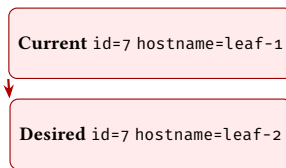


Figure 29. Diff engine flow. Two topologies are decomposed by layer, diffed by semantic key, and merged into a deterministic mutation plan containing six change types.

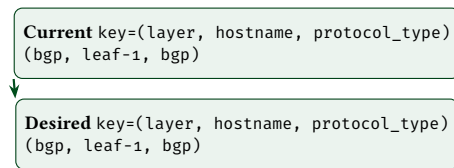
ID-based matching (fragile)



Insertion shifts IDs → false updates

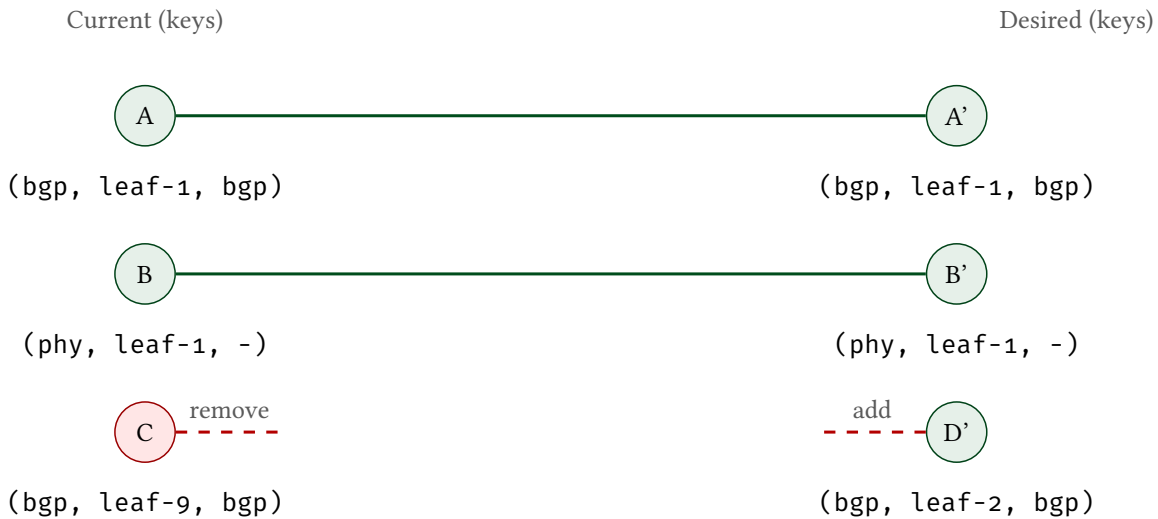
The diff engine compares nodes/edges by meaning (labels) rather than storage IDs, preventing cascading noise.

Semantic-key matching (stable)



Stable under inserts → true diffs only

Figure 30. Why semantic keys matter. Matching by integer ID turns benign insertions into widespread “updates”; matching by a composite semantic key isolates changes to the nodes that actually changed.



Diffing becomes a key-alignment problem: matched keys compare properties; unmatched keys become additions/removals (no ID noise).

Figure 31. Diffing as semantic key alignment. Nodes are matched by composite meaning (layer, hostname, protocol_type), then compared field-by-field; unmatched keys yield clean add/remove operations instead of cascaded updates.

7.4 AST Transformations (TSDM Layer)

The **Target-Specific Device Model (TSDM)** layer implements the compiler’s lowering phase via DSL-driven **AST-to-AST transformations**, whose blueprint syntax is described in §5.9. This section details the execution engine that bridges logical design and physical hardware constraints.

7.4.1 The Transformation DSL

Lowering rules are defined in the blueprint using an expressive rewrite-rule grammar. Each rule consists of a `match_expr` (CEL predicate) and an `apply` map (field mutations).

Example: NX-OS interface lowering

```
transforms:
- name: "nxos_lowering"
  when: "device_os == 'nxos'"
  rules:
  - match_expr: "kind == 'interface' && name.startsWith('Ethernet')"
    apply:
      name: "name + '/1'" # Ethernet1 -> Ethernet1/1
      mtu: "9216"         # Force jumbo frames
```

7.4.2 Execution Logic

The transformation engine performs a deep-walk of the DeviceIR stanzas. For each stanza, it:

1. Binds the stanza's fields as CEL variables.
2. Evaluates the `match_expr` predicate.
3. If matched, executes the `apply` expressions to mutate the stanza's state.

This structured approach allows complex hardware logic—such as mapping logical ports to specific line-card slots in a modular chassis—to be expressed as programmable transformations rather than hardcoded logic or brittle templates.

The key architectural insight is that `DeviceContext` accumulates state across the entire primitive pipeline:

1. `mesh_nodes`: writes `mesh_type`, `peer_count`, and optionally `peer_interfaces` into metadata.
2. `provision_ips`: writes `src_ip`, `dst_ip`, `subnet`, `vrf`.
3. `build_protocol_layer`: deep-merges all of the above into the cloned protocol node, plus the config overlay.
4. `allocate_resources`: writes the allocated resource value (e.g., `asn`) into metadata.

By the time Stage 4 (rendering) occurs, each node's `DeviceContext` carries hostname, IPs, protocol metadata, interface assignments, and any custom fields—the complete information needed to produce a device configuration file.

8 CLI Reference

NETCFG provides twelve CLI commands spanning the development-to-deployment lifecycle. All commands use `clap` for argument parsing. Table 35 shows which pipeline layers each command invokes.

Table 35. CLI commands and the pipeline layers they invoke. ✓ = executes; ◦ = optional (via flag).

Command	<i>L1 Intent</i>	<i>L2 Assert</i>	<i>L3 Mapping</i>	<i>L4 TSDM</i>	<i>L5 Render</i>	Primary output
generate	✓	✓	✓	✓	✓	Device config files
validate	✓	✓			◦	Pass/fail exit code
plan	✓	✓			◦	Mutation diff
inspect	◦					AST / DOT / trace
ci-run	✓	✓	✓	✓	✓	JSON report
lint	✓	✓				Assertion results
impact	✓	✓				Risk assessment
report	✓	✓				Markdown report
fmt						Formatted YAML
import						.nte archive
export-clab	✓	✓				clab.yaml
sync						.nte archive / stdout

8.1 netcfg generate

Runs the full five-layer compilation pipeline (§6), writing artefacts to disk.

```
netcfg generate <topology> <blueprint> \
  --templates-dir <DIR> \
  [--output-dir <DIR>] [--emit-ir] [--verbose]
```

Argument	Default	Description
topology	—	Path to annotated topology archive (.nte) or NSoT URI
blueprint	—	Path to blueprint YAML
--templates-dir	—	Directory containing .j2 templates
--output-dir	./output	Output directory for generated files
--emit-ir	false	Also write {hostname}.ir.json
--verbose	false	Print per-node rendering progress

Output: one .cfg file per device (plus .ir.json if --emit-ir). Absolute paths are printed to stdout, one per line, suitable for piping.

8.2 netcfg validate

Executes the pipeline in dry-run mode: all primitives run with `dry_run=true`, skipping DataFrame writes but verifying what *would* change. Optionally pre-flights templates by rendering to memory.

```
netcfg validate <topology> <blueprint> \
  [--templates-dir <DIR>] [--verbose]
```

Exit codes: 0 on success, non-zero on validation failure (parse errors, assertion failures, template syntax errors, IP pool conflicts).

8.3 netcfg plan

Computes the topology `MutationPlan` (current vs. desired) and displays additions, removals, and property changes. With `-diff`, it renders before/after configs and shows a unified text diff per device using the `similar` crate.

```
netcfg plan <topology> <blueprint> \
  [--diff] [--templates-dir <DIR>] [--verbose]
```

Output without `-diff`: colourised plan (green for additions, red for removals, yellow for modifications). Summary line: +N additions, -N removals across M layers.

Output with `-diff`: unified diff per device, preceded by `diff -cfg {hostname}`.

8.4 netcfg inspect

Provides three views into intermediate state:

```
netcfg inspect <path> --ast          # Print blueprint AST
netcfg inspect <path> --topology      # Export as Graphviz DOT
  [--blueprint <FILE>]
netcfg inspect <path> --trace <NODE_ID> # Trace node state changes
  --blueprint <FILE>
```

- `--ast`: Parses the file as a blueprint and prints the AST tree.
- `--topology`: Exports the graph as Graphviz DOT. With `--blueprint`, applies the blueprint first. Output includes per-layer subgraph clustering, hostname labels, and subnet annotations.
- `--trace <node_id>`: Replays primitive execution and prints before/after DeviceContext snapshots for the specified node at each primitive step. When blueprints fail with selector or assertion errors, finding the root cause in a 1,000-node graph is challenging; `--trace` is the primary debugging workflow for blueprint authors.

8.5 netcfg ci-run

The CI/CD integration point. Runs validate, plan, and config diff in a single invocation.

```
netcfg ci-run <topology> <blueprint> \  
  --templates-dir <DIR> [--json]
```

With `--json`, emits a structured payload:

```
{  
  "status": "success",  
  "topology_additions": 42,  
  "topology_removals": 0,  
  "configuration_diffs": {  
    "core-0": "- ip address 10.0.0.4/30\n+ ip address 10.0.0.8/30"  
  },  
  "warnings": []  
}
```

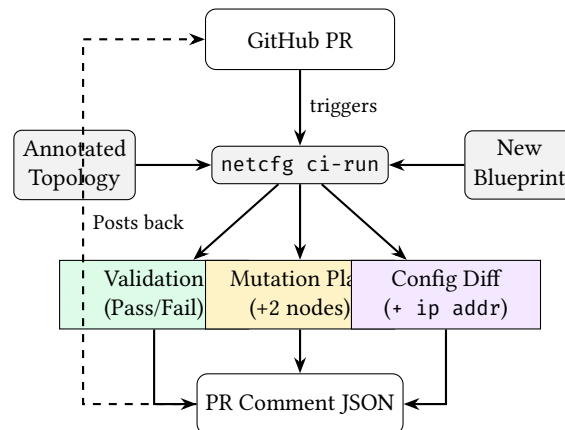


Figure 32. The NetDevOps CI/CD pipeline enabled by `ci-run`.

This integrates with GitHub Actions, GitLab CI, or any CI system that can parse JSON.

8.6 netcfg import

Imports a YAML topology definition into an NTE archive.

```
netcfg import <yaml_path> <output.ntc>
```

The importer reads a topogen-format YAML file (defining nodes with types, layers, and properties) and serialises it to an `.ntc` archive that other commands consume.

8.7 netcfg lint

Runs design-rule assertions without the full validation pipeline. Applies the blueprint in dry-run mode, then evaluates assertions and reports results.

```
netcfg lint <topology> <blueprint> [--verbose]
```

Returns assertion results with pass/fail status, severity, and diagnostic messages. Exit code 1 only when error-severity assertions fail; warnings are reported but do not cause failure.

8.8 netcfg fmt

Standardises blueprint YAML formatting. Parses the blueprint, re-serialises it in canonical form, and compares byte-for-byte. Idempotent.

```
netcfg fmt <blueprint> [--check] [--verbose]
```

With `--check`, exits with code 1 if formatting changes are needed (useful as a pre-commit hook or CI gate). Without `--check`, overwrites the file in place.

8.9 netcfg impact

Calculates the operational blast radius of a blueprint change against a base topology. The topology argument accepts either a local `.nte` archive path or an NSoT URI (e.g. `netbox://host?site=ldn1`); see `netcfg sync` (§8.12) for URI syntax.

```
netcfg impact <topology> <blueprint> [--detail] [--verbose]
```

The `--detail` flag expands the low-risk section to list individual nodes (by default only a count is shown).

For each mutation in the computed plan, assigns a risk level:

Risk	Colour	Trigger
High	Red	Node removal; critical property change (asn, router_id, src_ip, subnet); physical link add/remove
Medium	Yellow	Node addition; logical link add/remove
Low	Green	Metadata-only update

Critical properties are: `asn`, `router_id`, `src_ip`, `dst_ip`, `subnet`, `loopback`, `mgmt_ip`. Output includes per-node risk attribution with reasons.

8.10 netcfg report

Generates an automated Markdown architecture report from a compiled topology.

```
netcfg report <topology> <blueprint> -o <output.md> [--verbose]
```

The report contains three sections:

1. **Node inventory:** hostname, type, OS, ASN, loopback, management IP.
2. **Physical links & IP addressing:** source/destination hosts, interfaces, IPs, subnets.
3. **Topology visualisation:** A `netviz-json` code block with a custom JSON schema (`NetVizNode`, `NetVizEdge`, `NetVizTopology`) suitable for rendering by external visualisation tools.

8.11 netcfg export-clab

Generates a Containerlab `clab.yaml` simulation file from a compiled topology, enabling rapid lab deployment for testing (e.g. the three-site mesh from §9). The topology argument accepts a local `.nte` archive or an NSoT URI (§8.12).

```
netcfg export-clab <topology> <blueprint> \  
-o <clab.yaml> [--configs-dir <DIR>] [--verbose]
```

Heuristically maps `device_os` to Containerlab container kinds: `cisco_iosxr` → `vr-xrv9k`, `nokia_srl` → `srl`, `arista_eos` → `ceos`, `default` → `linux`. When `--configs-dir` is provided, bind-mounts generated configuration files into OS-specific remote paths (e.g. `/etc/opt/srlinux/config.json` for SRL, `/mnt/flash/startup-config` for cEOS).

8.12 netcfg sync

Synchronises topology data with external Network Source of Truth (NSoT) systems. Two subcommands are available:

```
netcfg sync pull <uri> [-o <output.nte>] [--token <TOKEN>] [--insecure]  
netcfg sync push <uri> <source> [--token <TOKEN>]
```

`sync pull` fetches a live topology from an NSoT and converts it into an NTE topology archive. The URI scheme identifies the provider: `netbox://host[:port]?site=X®ion=Y` targets a NetBox [6] instance

via its GraphQL endpoint; query parameters are passed as filters. Authentication is resolved in order: the `-token` flag, then the `NETBOX_TOKEN` environment variable. When `-o` is supplied the topology is persisted to disk; otherwise a summary (node and edge counts) is printed. The `-insecure` flag disables TLS certificate verification for self-signed development instances.

`sync push` is a planned stub that will write intent back to the NSoT (see §13).

Note: NSoT URI in other commands

The topology argument of `impact` (§8.9) and `export-clab` (§8.11) also accepts a `netbox://` URI. Internally these commands dispatch through the same `load_topology_source()` helper, so a live NSoT topology can be used without a local archive.

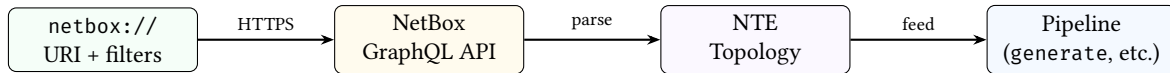


Figure 33. NSoT synchronisation data flow.

9 End-to-End Worked Example

We trace the `three-site-mesh.yaml` blueprint through every pipeline stage. The six stages below correspond to the five pipeline layers in §6: Stages 1–3 are subsumed by Layer 1 (Intent Transformation), Stage 4 maps to Layer 3 (DeviceIR), Stage 5 to Layer 5 (Serialisation), and Stage 6 is the atomic write that concludes Layer 5.

The annotated topology has 12 routers across 3 sites (4 per site: `site-a-r1` through `site-c-r4`), each annotated with `role`, `site`, and `device_os` properties that the blueprint’s selectors reference.

9.1 Input: Blueprint

Listing 12. `three-site-mesh.yaml` (abridged).

```

version: 1
layers:
  # Stage 1: Full mesh within each site
  - name: input
    primitives:
      - type: mesh_nodes
        selector: "nodes[site='a']"
        mesh_type: full
      - type: mesh_nodes
        selector: "nodes[site='b']"
        mesh_type: full
      - type: mesh_nodes
        selector: "nodes[site='c']"
        mesh_type: full

  # Stage 2: IP allocation per site
  - name: input
    primitives:
      - type: provision_ips
        selector: "edges[src>=0]"
        pool: "10.1.0.0/24"
        subnet_size: 30
        strategy: dense

  # Stage 3: BGP overlay
  - name: input
    primitives:
      - type: build_protocol_layer
        selector: "nodes[true]"
        layer: bgp
        config:
          protocol_type: bgp
          asn_base: "65000"
  
```

9.2 Stage 1: Parse and validate

The parser validates the YAML, confirms version: 1, and compiles each selector string (`nodes[site='a']`, `edges[src>=0]`, etc.) to a CEL program. The normaliser converts `site='a'` to `site=="a"`.

9.3 Stage 2: Shadow topology

The executor clones the annotated topology via a serialisation round-trip, creating an isolated shadow copy. All subsequent mutations operate on the shadow; the source topology is never modified.

9.4 Stage 3: Primitive execution

mesh_nodes (three invocations): Creates $3 \times \binom{4}{2} = 18$ edges (6 per site). The selector `nodes[site='a']` matches the 4 nodes with `{site: "a"}` in their `data_json`. Each node gets `mesh_type: "full"` and `peer_count: 3` written to its metadata.

provision_ips: The selector `edges[src>=0]` matches all 18 edges. Edges are sorted by (*src, dst*). The five-pass algorithm allocates /30 subnets from `10.1.0.0/24`:

- Edge (1,2): subnet `10.1.0.0/30`, IPs `10.1.0.1` and `10.1.0.2`
- Edge (1,3): subnet `10.1.0.4/30`, IPs `10.1.0.5` and `10.1.0.6`
- ...continuing through all 18 edges.

After this stage, node `site-a-r1` has: `src_ip: "10.1.0.1"`, `subnet: "10.1.0.0/30"`, `vrf: "default"`.

build_protocol_layer: Clones all 12 physical nodes into the bgp layer (new IDs 13–24). Each clone receives:

- `protocol_type: "bgp"` and `asn_base: "65000"` (from the config overlay)
- `_source_id: 1` (parent mapping)
- All properties from the physical node, including the IPs written by `provision_ips` (via deep merge)

9.5 Stage 4: Mapping evaluation

The companion mapping (`three-site-mesh-mapping.yaml`) extracts per-device stanzas from the enriched graph:

```
rules:
- selector: "edges[layer=='physical']"
  stanza:
    kind: "interface"
    fields:
      name: "{{ node.peer_interfaces[peer.hostname] }}"
      description: "Link to {{ peer.hostname }}"
      ip_address: "{{ node.src_ip }}"
      mtu: 9000
- selector: "nodes[layer=='bgp']"
  stanza:
    kind: "bgp"
    fields:
      local_asn: "{{ node.asn }}"
      router_id: "{{ node.loopback }}"
      neighbors: "{{ node.bgp_peers }}"
```

The evaluator walks each rule's selector, matches the relevant nodes/edges, and emits one stanza per match. Each stanza carries provenance: `{rule_id: "rule_0"}`, linking rendered output back to its source rule.

9.6 Stage 5: Render

Stanzas are grouped by hostname. The Jinja2 template iterates over each node's stanza list and emits vendor-specific CLI syntax. For `site-a-r1`, the rendered output includes:

```
! site-a-r1.conf (abridged)
interface Ethernet0
  description Link to site-a-r2
  ip address 10.1.0.1/30
  mtu 9000
!
interface Ethernet1
  description Link to site-a-r3
  ip address 10.1.0.5/30
  mtu 9000
```



```
!
router bgp 65001
  router-id 10.255.0.1
  neighbor 10.1.0.2 remote-as 65001
  neighbor 10.1.0.6 remote-as 65001
```

Each of the 12 routers receives a distinct configuration derived from its position in the topology graph—no two devices share the same output.

9.7 Stage 6: Atomic write

The TransactionalOutput writer collects all files in temporary locations, then atomically renames them via POSIX `rename(2)`. Per device, three files are written:

- `site-a-r1.conf`: the rendered configuration
- `site-a-r1.deviceir.json`: the raw stanza list (audit trail)
- `site-a-r1.lineage.json`: per-line provenance mapping

If any stage fails, all temporary files are dropped—no partial output directory is ever visible.

9.8 Case Study: Compliance-Hardened Financial Network

The preceding example demonstrates basic topology and IP allocation. This second worked example—drawn from `examples/case_studies/10_financial_compliance.yaml` (387 lines)—exercises the compiler’s full primitive breadth across a PCI-DSS/SOX-compliant financial trading network.

Scenario. A financial institution operates six devices: two core routers (OSPF backbone, route reflectors), one DMZ router (internet-facing, heavily filtered), two trading-floor switches (low-latency, MACsec encrypted), and one out-of-band management node (TACACS+/RADIUS). The blueprint must enforce: MACsec on all trading links, zone policies blocking DMZ-to-trading traffic, authenticated NTP and SNMP v3 for compliance, and strict blast radius controls (maximum one deletion, maximum 5% change).

Blueprint structure. The blueprint declares nine layers, each with an explicit `requires` dependency:

```
layers:
- name: topology           # 4 mesh primitives
  requires: []
- name: addressing         # 3 provision_ips primitives
  requires: [topology]
- name: routing            # OSPF + BFD
  requires: [addressing]
- name: bgp_safety         # Bogon/RFC1918 filters on DMZ
  requires: [routing]
- name: encryption         # MACsec AES-256-GCM on trading links
  requires: [bgp_safety]
- name: zone_security      # DMZ->trading deny; mgmt->all SSH-only
  requires: [encryption]
- name: aaa                # TACACS+ / RADIUS / 802.1X
  requires: [zone_security]
- name: compliance_services # NTP, syslog-TLS, SNMPv3, NetFlow
  requires: [aaa]
- name: secrets            # inject_secrets for all keys
  requires: [compliance_services]
```

Layer dependency trace. The compiler resolves this as a linear chain: `topology` → `addressing` → `routing` → `bgp_safety` → `encryption` → `zone_security` → `aaa` → `compliance_services` → `secrets`. Each layer’s primitives see the accumulated state from all preceding layers—for example, the `build_zone_policy` primitives in `zone_security` can reference IP addresses assigned two layers earlier. This progressive enrichment is the same pattern described in §2.5, but exercised across nine layers rather than three.

Assertions. The blueprint declares nine assertions spanning topology, security, and compliance concerns:

Table 36. Assertions in the financial compliance case study.

Assertion	Type	Purpose
core-connected	is_connected	Network is not partitioned
dmz-reaches-core	reachability	DMZ can reach core routers
trading-reaches-core	reachability	Trading switches reach core
zone-policy-dmz-trading	zone_policy_exists	PCI-DSS: DMZ→trading blocked
ntp-compliance	use_rule	PCI-DSS 10.4: synchronised clocks
syslog-compliance	use_rule	PCI-DSS 10.5: centralised logging
all-have-hostname	field_exists	All routers have hostname
inventory-schema	match_schema	SOX: serial, firmware, site required
hostnames-unique-per-site	unique_per_group	No duplicate hostnames per site

Key observations.

- **Blast radius control.** The `blast_radius` section caps deletions at one node and total changes at 5%—reflecting a change-averse financial environment. The diff engine (§7.3) enforces these limits before any mutation is applied.
- **Abstraction benefit.** Twenty-one primitives across nine layers produce per-device configurations for six devices: OSPF adjacencies, BFD timers, MACsec key chains, zone-based firewall rules, TACACS+ command authorisation, 802.1X port security, NTP with authentication, syslog over TLS, SNMP v3 authPriv, and NetFlow/IPFIX collectors. Expressing this manually per device would require several hundred lines of vendor CLI; the blueprint captures the *intent* in 387 lines of YAML.
- **Secret injection.** The final secrets layer uses `inject_secrets` to bind environment variables (`${TACACS_KEY}`, `${MACSEC_CAK}`, etc.) at compile time, keeping credentials out of the blueprint source.

10 Error Model

NETCFG uses typed error enums with `thiserror` derivation and `miette` diagnostic annotations. Table 37 summarises the error types.

Table 37. Error enums and their variants.

Enum	Variants	Context
<code>ParseError</code>	<code>YamlSyntax</code> , <code>Validation</code> , <code>Deserialize</code> , <code>LayerOrdering</code>	Blueprint parsing
<code>SelectorError</code>	<code>InvalidSyntax</code> , <code>Compile</code> , <code>Evaluate</code> , <code>NonBooleanResult</code> , <code>Polars</code> , <code>Json</code> , <code>TopologyAccess</code>	Selector compilation and evaluation
<code>PrimitiveError</code>	<code>Validation</code> , <code>Execute</code> , <code>Selector</code> , <code>Topology</code> , <code>Graph</code> , <code>Polars</code> , <code>Json</code>	Primitive execution
<code>ExecuteError</code>	<code>Parse</code> , <code>Primitive</code> , <code>Shadow</code> , <code>Plan</code> , <code>Assertion</code>	Top-level executor
<code>ShadowError</code>	<code>Clone</code> , <code>Commit</code> , <code>Validation</code>	Shadow topology lifecycle
<code>AssertionError</code>	<code>Failed</code> , <code>Selector</code>	Assertion evaluation
<code>DiffError</code>	<code>TopologyAccess</code> , <code>ParseError</code>	Configuration diffing
<code>RenderError</code>	<code>Io</code> , <code>Jinja</code> , <code>Json</code> , <code>MissingTemplate</code>	Template rendering

`ExecuteError` variants carry `miette::Diagnostic` annotations with diagnostic codes (e.g., `netcfg::parse`, `netcfg::primitive::execution_failed`, `netcfg::assertion`) enabling structured error handling in CI pipelines.

`ParseError::YamlSyntax` includes a source span (`miette::SourceSpan`) pointing at the offending line, enabling IDE-quality error rendering.

11 Editor and Service Integration

11.1 Language Server (LSP)

The `netcfg-lsp` crate provides a Language Server Protocol implementation built on `tower-lsp` and Tokio. It communicates via stdin/stdout and offers three capabilities:

1. **Real-time diagnostics.** On file open, change, and save, the server parses the blueprint and maps errors to LSP diagnostics with line-column precision. YAML parse errors (which embed location as free text) are converted to line/column positions via pattern matching. Diagnostics are published with source label `ank_netcfg` and severity `ERROR`.
2. **Primitive auto-completion.** When the cursor follows a `type:` token (detected by a suffix heuristic), the server offers completion items for all twenty-three primitives, each with a one-line docstring. Trigger characters are space and colon.
3. **Document synchronisation.** Open documents are tracked in a `DashMap<String, String>` (lock-free concurrent map). Full-text sync (`TextDocumentSyncKind::FULL`) is used for correctness. On close, the document is removed and diagnostics are cleared.

11.2 REST API Microservice

The `netcfg-api` crate exposes the compiler as a stateless HTTP service via `axum`, enabling CI/CD pipelines and enterprise orchestrators to compile blueprints without a local `netcfg` binary.

Endpoint: POST `/compile` (port 3000).

Table 38. Compile endpoint request and response schema.

Direction	Field	Type	Description
Request	<code>topology_zip_base64</code>	String	Base64-encoded <code>.nnt</code> archive
Request	<code>blueprint_yaml</code>	String	Raw YAML blueprint
Response	<code>status</code>	String	<code>success</code> or <code>error</code>
Response	<code>configs</code>	<code>Map<String, String></code>	Hostname → rendered config
Response	<code>device_ir</code>	<code>Map<String, JSON></code>	Hostname → DeviceContext IR
Response	<code>warnings</code>	<code>Vec<String></code>	Compilation warnings

The service executes in a strict sandbox mode: `inject_secrets` and `wasm_plugin` primitives are disabled by default to prevent remote exfiltration of environment variables. The compilation pipeline decodes the base64 payload into a temporary file, loads the topology, parses the blueprint, executes via `Executor::apply()`, and extracts per-device IR. All processing is in-memory with no persistent state; the service scales horizontally behind a load balancer.

12 Evaluation

This section evaluates `NETCFG` along three axes: expressiveness (case study corpus), abstraction leverage (blueprint-to-output ratio), and performance (micro-benchmarks on synthetic topologies).

12.1 Case study corpus

Table 39 summarises ten case studies spanning ISP peering, campus compliance, WAN migration, data-centre multi-tenancy, service provider MPLS, campus NAC, SD-WAN IPsec, multicast video distribution, zero-trust data-centre, and financial compliance.

Table 39. Case study corpus. Each blueprint was compiled successfully with all assertions passing.

#	Case study	Lines	Layers	Prims	Asserts
1	ISP dual-stack peering	166	6	13	5
2	Campus compliance audit	153	2	2	8
3	WAN migration	183	6	9	4
4	Multi-tenant DC	210	4	7	8
5	SP MPLS core	220	6	15	5
6	Campus NAC	292	7	20	6
7	SD-WAN IPsec	204	6	12	5
8	Multicast video	251	7	18	5
9	Zero-trust DC	331	7	19	5
10	Financial compliance	387	9	21	9
Total / Mean		2,397	6.0	13.6	6.0

The corpus covers 136 primitives across 60 layers, exercising all twenty-three primitive types. Mean blueprint length is 240 lines of YAML, yet each blueprint generates per-device configurations for between 4 and 20 devices. The assertion count ranges from 2 (campus compliance, which relies on eight schema-validation assertions) to 9 (financial compliance, which encodes PCI-DSS and SOX controls).

12.2 Abstraction ratio

The *abstraction ratio* measures how many lines of vendor-specific output one line of blueprint produces. A ratio above 1.0 indicates that the blueprint is more concise than the output it generates; higher ratios indicate greater abstraction leverage.

For a typical deployment of the ISP dual-stack peering case study (case 1, 12 devices), the compiler produces approximately 2,400 lines of device configuration from 166 lines of blueprint—an abstraction ratio of $\approx 14.5:1$. The financial compliance case study (case 10, 6 devices) produces approximately 1,800 lines from 387 lines of blueprint ($\approx 4.7:1$). The lower ratio reflects the compliance blueprint’s heavier assertion and blast-radius specification, which are zero-cost at output time but contribute to blueprint line count.

12.3 Performance

All benchmarks use the Criterion statistical harness (`netcfg-core/benches/pipeline_benchmarks.rs`) on synthetic ring and hub-and-spoke topologies. Table 40 reports median times on an Apple M4 Max (macOS 15.3.2).

Table 40. Micro-benchmark results (median, Criterion).

Benchmark	Median	Notes
Selector eval (1K nodes)	6.5 ms	CEL hostname != "
Selector eval (10K nodes)	274 ms	CEL hostname != "
IPAM (1K edges)	48 ms	provision_ips, dense, /8 pool
IPAM (10K edges)	5.3 s	provision_ips, dense, /8 pool
Mesh nodes (10K, hub&spoke)	2.7 s	10 hubs + 9,990 spokes
Protocol layer (10K)	5.7 s	build_protocol_layer clone
Rendering (1 node)	1.9 μ s	Single-node MiniJinja
End-to-end (1K nodes)	53 ms	Full Executor::apply
End-to-end (10K nodes)	5.7 s	Full Executor::apply

Summary. The case study corpus demonstrates that NETCFG’s twenty-three primitives cover a wide range of production network designs, from simple campus topologies to complex multi-protocol financial environments. Abstraction ratios of 4.7–14.5:1 confirm that blueprint-level intent is substantially more concise than the equivalent device configuration. The micro-benchmarks show that a 1,000-node end-to-end compilation completes in 53 ms, and even at 10,000 nodes the pipeline completes in under 6 s—well within interactive-use expectations. Template rendering is sub-microsecond per node (1.9 μ s), confirming that the render layer adds negligible overhead.

13 Limitations

13.1 `edge_properties` not applied (`mesh_nodes`)

Description. The `MeshNodesSpec.edge_properties` field is accepted by the blueprint parser but intentionally not applied. Applying it requires an edge `DataFrame` API on `nte_topology::Topology` equivalent to `set_dataframe/update_dataframe` but keyed on (`source_id`, `target_id`) pairs.

Workaround. Edge properties can be encoded as node metadata on both endpoints (e.g., `link_bandwidth` as a per-node field keyed by peer ID).

Planned resolution. Awaiting `nte-topology` ≥ 0.2 edge `DataFrame` support. A tracking test (`edge_properties_deferred_until_nte_edge_dataframe`) is present in the test suite.

13.2 Mapping AST inspection not implemented

Description. The `inspect -ast` command can parse and display blueprint ASTs but does not yet support mapping documents. Attempting to inspect a mapping file prints “Mapping AST visualization coming soon.”

Workaround. Inspect the mapping YAML directly. The format is simple (a rules list with selector and stanza fields).

Planned resolution. When the mapping DSL grows beyond simple selector/stanza pairs, AST visualisation will be added.

13.3 Shadow validation is a placeholder

Description. The `ShadowTopology::validate()` method currently only checks that the shadow did not become empty. It does not perform structural integrity checks (dangling edges, orphaned protocol nodes, etc.).

Workaround. Design-rule assertions can catch many structural problems. Use `min_edges` and `field_exists` assertions as guardrails.

Planned resolution. Integrate with the NTE policy validation framework when available.

13.4 No intra-primitive rollback

Description. If a primitive fails partway through execution (e.g., `provision_ips` exhausts its pool after allocating half the edges), the shadow topology retains the partial mutations. The pipeline aborts (returning an error), and the shadow is discarded, but there is no mechanism to roll back individual primitives within a shadow.

Workaround. This is usually not a problem: the entire shadow is discarded on error, so no partial state reaches production. The limitation matters only for debugging: the error message may not reflect the full extent of partial mutations.

Planned resolution. No current plan. Intra-primitive rollback would require snapshotting the shadow before each primitive, which doubles memory usage.

13.5 `generate CLI` bypasses mapping document

Description. The `generate CLI` command renders configs directly from `DeviceContext` via templates, bypassing the mapping document. The full pipeline (with mapping evaluation and `DeviceIR`) is implemented in `config_gen::generate_configs()` but the CLI does not expose it yet.

Workaround. Use the `generate_configs()` function directly from Rust code for the full pipeline with mapping support.

Planned resolution. Add `-mapping` flag to the `generate CLI` command.

13.6 No incremental compilation

Description. Every invocation recompiles the entire topology from scratch. The diff engine can identify which devices changed, but the pipeline cannot skip unchanged devices during compilation.

Workaround. Use `ci-run -json` to compute the diff, then selectively push only the changed configuration files.

Planned resolution. Under investigation. Incremental compilation requires caching intermediate state (the desired topology) between runs and detecting which blueprint changes invalidate which nodes.

13.7 No runtime verification or drift detection

Description. `NETCFG` is a generation-time tool. It produces configuration artifacts but does not push them to devices, monitor compliance, or detect post-deployment configuration drift. Closed-loop verification requires integration with external tools.

Workaround. Periodically regenerate configurations from the canonical blueprint and diff against deployed state using `ci-run -json`. Use Batfish [10] or gNMI-based telemetry for runtime compliance checking.

13.8 Shadow topology cloning cost

Description. The transactional shadow is created via a JSON serialisation round-trip, which is $O(n)$ in topology size. For 10,000-node topologies this adds ~200 ms of overhead.

Workaround. Not typically a bottleneck, but for interactive workflows the cost is noticeable. Future work may explore copy-on-write graph snapshots in the NTE engine.

13.9 WASM plugin is experimental

Description. The `wasm_plugin` primitive validates module compilation and symbol resolution but does not implement full linear memory I/O for JSON data exchange between the Rust host and WASM guest. The current implementation is proof-of-concept only.

Workaround. Use `custom_primitive` with nested built-in primitives for extensibility. For complex transformations, implement them as Rust primitives.

Planned resolution. Full C-ABI memory allocation between host and guest is required for production use. See §15.

13.10 Mapping document limitations

Description. Two limitations affect the mapping workflow: (1) the `inspect -ast` command does not yet support mapping documents (§13.2); and (2) there is no auto-generation of mapping rules from OpenConfig YANG schemas or device inventory data—operators must author the mapping document by hand, which requires knowledge of DeviceContext field names and stanza structure.

Workaround. The mapping format is intentionally simple (a rules list with selector and stanza fields). Example mappings ship with the worked examples (`three-site-mesh-mapping.yaml`, `evpn-vxlan-mapping.yaml`).

13.11 NSoT push not implemented (`netcfg sync push`)

Description. The `netcfg sync push` subcommand is defined in the CLI but returns an error when invoked. Pushing intent back to an NSoT system requires a write-capable provider interface (e.g. NetBox REST API mutations) that has not yet been implemented.

Workaround. Export the compiled topology to an NTE archive (`sync pull -o`) and ingest it into the NSoT manually.

Planned resolution. A `NetBoxWriter` provider will be added alongside the existing `NetBoxProvider` read path.

14 Related Work

Network configuration synthesis and verification have received significant attention. We position NETCFG against three strands of related work.

Configuration synthesis and formal methods. Early declarative networking languages like Frenetic [15] and NetKAT [16] pioneered the separation of intent from mechanism via formal semantics in SDN, but did not address legacy CLI-driven environments. Propane [17] compiles high-level routing policies written in a purpose-built language into per-device BGP configurations, proving correctness by construction. SyNET [18] uses constraint solving to synthesise router configurations from a formal network specification. Both systems target routing exclusively; NETCFG covers a broader surface (IP allocation, zone policies, NAT, overlays) at the cost of weaker formal guarantees—design-rule assertions rather than proofs. On the data plane, P4 [19] demonstrated that packet-processing hardware could be programmed via a compiler funnel that lowers abstract intent into target-specific architectures. NETCFG applies this exact compiler philosophy to the management plane.

Configuration analysis and verification. Header Space Analysis [20] introduced the foundation for statically checking network properties. Building on this, Batfish [21] models device configurations as a data plane and evaluates reachability queries offline; Minesweeper [22] extends this with symbolic model checking against network-wide invariants. Config2Spec [23] mines specifications from existing configurations to bootstrap verification. Forward Enterprise [24] provides commercial intent verification with a digital-twin approach. These tools analyse *existing* configurations, whereas NETCFG operates upstream: it *generates* the configurations that such tools would later verify, and its assertion framework provides lightweight invariant checking at compile time.

Operational tooling and orchestration. Ansible [8], Salt [25], and Nornir [9] are widely deployed for configuration deployment and orchestration. Cisco NSO [26] provides model-driven (YANG-based) service orchestration. netlab [27] generates lab topologies for testing. These tools focus on *deploying* or *simulating* configurations rather than synthesising them from intent. Frenetic [15] introduced a functional network programming language for SDN controllers, inspiring the declarative philosophy that NETCFG applies to traditional device configuration.

NETCFG occupies a middle ground: it is more expressive than pure routing synthesisers, less formally rigorous than model checkers, and operates at a higher abstraction level than deployment tools. Its graph-based compilation pipeline (§6) and pluggable primitive architecture (§5.5) are, to our knowledge, unique in this space.

15 Future Work

- **GPU-accelerated primitive execution:** Offloading selector evaluation and IPAM allocation to the GPU via NTE’s WebGPU compute backend, targeting 100K+ node topologies.
- **WASM linear memory I/O:** The current `wasm_plugin` primitive validates module compilation and symbol resolution but does not implement full linear memory I/O for JSON data exchange. Full C-ABI memory allocation between the Rust host and WASM guest is required for production use.
- **API template rendering:** The `POST /compile` endpoint currently returns placeholder configuration text; integrating the MiniJinja rendering engine into the API service is planned.
- **Mapping CLI integration:** Exposing the full pipeline (including mapping evaluation) through the `generate` CLI command.
- **Incremental compilation:** Caching intermediate state to avoid recompiling unchanged portions of the topology.
- **LSP assertion integration:** Extending the language server to evaluate design-rule assertions in real time, providing inline warnings for selector violations and IP pool capacity.
- **Annotation defaults and inheritance:** Currently, every node must carry its own annotations (e.g. `device_os: eos` on each device). A default/inheritance mechanism—for example, site-level or group-level defaults that individual nodes can override—would reduce repetition in large topologies and reinforce the “topology as source code” model.

References

- [1] S. Shenker, “The future of networking, and the past of protocols,” in *Open Networking Summit*, vol. 20, Palo Alto, CA: Open Networking Foundation, 2011, pp. 1–30.
- [2] P. Gill, N. Jain, and N. Nagappan, “Understanding network failures in data centers: Measurement, analysis, and implications,” in *ACM SIGCOMM computer communication review*, vol. 41, New York, NY, USA: ACM, 2011, pp. 350–361.
- [3] J. Zhu et al., “MALT: Multi-abstraction layer topology,” in *17th USENIX Symposium on Networked Systems Design and Implementation (NSDI 20)*, Santa Clara, CA: USENIX Association, 2020, pp. 313–328.
- [4] OpenConfig Working Group, *OpenConfig*, <https://openconfig.net>, 2023.
- [5] M. Bjorklund, “The YANG 1.1 data modeling language,” IETF, RFC 7950, 2016.
- [6] NetBox Community, *NetBox: The source of truth for network automation*, <https://netbox.dev/>, 2023.
- [7] Network to Code, *Nautobot: Network source of truth and automation platform*, <https://nautobot.com/>, 2023.
- [8] Red Hat, *Ansible automation platform*, <https://www.ansible.com/>, 2023.
- [9] Nornir Automation, *Nornir: Pluggable multi-threaded framework with inventory management to help operate collections of devices*, <https://nornir.tech/>, 2023.
- [10] Intentionet, *Batfish: Network configuration analysis tool*, <https://www.batfish.org/>, 2023.
- [11] C. Lattner and V. Adve, “LLVM: A compilation framework for lifelong program analysis & transformation,” in *International Symposium on Code Generation and Optimization, 2004. CGO 2004.*, IEEE, San Jose, CA: IEEE, 2004, pp. 75–86.
- [12] Google, *Common Expression Language (CEL)*, <https://github.com/google/cel-spec>, 2023.
- [13] S. Knight, *AutoNetKit NTE: The network topology engine*, 2026.
- [14] Bytecode Alliance, *Wasmtime: A fast and secure runtime for WebAssembly*, <https://wasmtime.dev/>, 2024.
- [15] N. Foster et al., “Frenetic: A network programming language,” in *Proceedings of the 16th ACM SIGPLAN International Conference on Functional Programming*, 2011, pp. 279–291.
- [16] C. J. Anderson et al., “NetKAT: Semantic foundations for networks,” in *Proceedings of the 41st ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, 2014, pp. 113–126.
- [17] R. Beckett, R. Mahajan, T. Millstein, J. Padhye, and D. Walker, “Propane: A general approach to network configuration synthesis,” in *Proceedings of the 2016 ACM SIGCOMM Conference*, 2016, pp. 440–453.
- [18] A. El-Hassany et al., “Network configuration synthesis with logic-based routing models,” in *Proceedings of the ACM on Programming Languages (OOPSLA)*, 2017.
- [19] P. Bosshart et al., “P4: Programming protocol-independent packet processors,” in *ACM SIGCOMM Computer Communication Review*, vol. 44, ACM New York, NY, USA, 2014, pp. 87–95.
- [20] P. Kazemian, G. Varghese, and N. McKeown, “Header space analysis: Static checking for networks,” in *9th USENIX Symposium on Networked Systems Design and Implementation (NSDI 12)*, 2012, pp. 113–126.
- [21] A. Fogel et al., “A general approach to network configuration analysis,” in *12th USENIX Symposium on Networked Systems Design and Implementation (NSDI 15)*, 2015, pp. 469–483.
- [22] R. Beckett, A. Gupta, R. Mahajan, and D. Walker, “A general approach to network configuration verification,” in *Proceedings of the 2017 Conference of the ACM Special Interest Group on Data Communication*, 2017, pp. 155–168.
- [23] R. Birkner et al., “Config2spec: Mining network specifications from network configurations,” in *17th USENIX Symposium on Networked Systems Design and Implementation (NSDI 20)*, 2020.
- [24] Forward Networks, *Forward Enterprise*, <https://www.forwardnetworks.com/>, 2024.
- [25] SaltStack, *Salt: Configuration management and orchestration*, <https://saltproject.io/>, 2024.
- [26] Cisco Systems, *Cisco Network Services Orchestrator (NSO)*, <https://www.cisco.com/c/en/us/products/cloud-systems-management/network-services-orchestrator/index.html>, 2023.
- [27] I. Pepelnjak, *Netlab: Network configuration and testing lab*, <https://netlab.tools/>, 2024.

A Blueprint YAML Schema Reference

Listing 13. Complete blueprint schema (annotated).

```
version: 1                                # integer, >= 1
imports: [<path>, ...]                   # optional, relative file paths to merge

# -- Named groups --
groups:
  <name>: <cel_selector>                 # shorthand: just a selector string
  # OR full form:
  <name>:
    selector: <cel_selector>
    kind: security_zone                  # optional
```



```

trust_level: <i32>          # required when kind is security_zone
interface_selector: <cel_selector> # optional, edge-level zone membership

# -- Named objects --
objects:
  addresses:
    <name>:
      prefixes: [<cidr>, ...]      # IPv4 CIDRs
      prefixes_v6: [<cidr>, ...]   # optional IPv6 CIDRs
  services:
    <name>:
      entries:
        - protocol: <string>      # e.g. "tcp", "udp"
          port: <u16>

# -- Named rules (CEL macros) --
rules:
  <name>: <cel_expression>      # referenced via use_rule check

layers:
  - name: <string>              # non-empty, layer identifier
    requires: [<string>, ...]   # optional, previously-declared layers
  primitives:

    # --- Topology tier ---

    # --- mesh_nodes ---
    - type: mesh_nodes
      selector: <cel_selector>
      mesh_type: full | hub_and_spoke
      hub_selector: <cel_selector> # required for h&s
      spoke_selector: <cel_selector> # required for h&s

    # --- build_protocol_layer ---
    - type: build_protocol_layer
      selector: <cel_selector>
      layer: <string>              # target layer
      config: <json_object>       # deep-merged into clone
      clone_underlying: <bool>    # default: false

    # --- Allocation tier ---

    # --- provision_ips ---
    - type: provision_ips
      selector: <cel_selector>
      pool: <cidr_or_pool_name>
      subnet_size: <u8>            # default: 30 (v4), 126 (v6)
      ipv6_pool: <cidr_or_pool_name> # optional dual-stack
      ipv6_subnet_size: <u8>
      strategy: dense | sparse | hash # default: dense
      pool_size: <u8>              # for global pool carving
      vrf: <string>                # default: "default"

    # --- allocate_resources ---
    - type: allocate_resources
      selector: <cel_selector>
      resource_type: <string>
      pool: <range_or_pool_name>
      strategy: dense              # default: dense
      pool_size: <u8>

    # --- global_pool ---
    - type: global_pool
      name: <string>
      pool: <cidr>
      vrf: <string>

    # --- global_resource_pool ---
    - type: global_resource_pool

```

```

name: <string>
resource_type: <string>
pool: <range>

# --- Policy tier ---

# --- build_routing_policy ---
- type: build_routing_policy
  selector: <cel_selector>
  policy_name: <string>
  statements:
    - name: <string> # sequence label
      action: permit | deny
      match_prefix_list: <string> # optional
      match_community_list: <string> # optional
      match_as_path: <string> # optional
      set_local_preference: <u32> # optional
      set_metric: <u32> # optional
      set_as_path_prepend: <string> # optional
      set_community: <string> # optional
      next_hop: <ip_address> # optional

# --- build_access_policy ---
- type: build_access_policy
  selector: <cel_selector>
  policy_name: <string>
  rules:
    - name: <string>
      action: permit | deny
      source_prefix: <cidr> # optional (IpNet)
      destination_prefix: <cidr> # optional (IpNet)
      protocol: <string> # optional
      source_port: <u16> # optional (integer)
      destination_port: <u16> # optional (integer)

# --- build_zone_policy ---
- type: build_zone_policy
  selector: <cel_selector>
  from_zone: <zone_name>
  to_zone: <zone_name>
  default_action: permit | deny # default: deny
  rules:
    - name: <string>
      action: permit | deny
      match: # optional (omit = match-all)
        source: <address_object> # optional
        destination: <address_object> # optional
        service: <service_object> # optional
        protocol: <string> # optional
        destination_port: <u16> # optional
        icmp_type: <string> # optional
      log: <bool> # optional
      stateful: <bool> # optional

# --- build_nat_policy ---
- type: build_nat_policy
  selector: <cel_selector>
  rules:
    - name: <string>
      type: snat | dnat | nat64 | nptv6
      match: # optional
        source_prefix: <cidr> # optional
        destination_prefix: <cidr> # optional
        protocol: <string> # optional
      snat: # when type: snat
        pool: <cidr>
      dnat: # when type: dnat
        destination: <ip>
        internal_port: <u16> # optional

```

```

    nat64:
        nat64_prefix: <string> # when type: nat64 # optional
    nptv6:
        internal_prefix: <string> # when type: nptv6 # optional
        external_prefix: <string> # optional
        allow_best_effort: <bool> # default: false
        from_zone: <zone_name> # optional
        to_zone: <zone_name> # optional
        log: <bool> # optional

# --- build_qos_policy ---
- type: build_qos_policy
  selector: <cel_selector>
  policies:
    <policy_name>:
      traffic_classes:
        <class_name>:
          classifier: {dscp: [...]} # optional
          marker: {dscp: <string>} # optional
          scheduler: # optional
            priority: <u32>
            bandwidth_percent: <u32>
          policer: # optional
            committed_rate: <string>
            burst_size: <string>
            exceed_action: <string>
          shaper: # optional
            peak_rate: <string>
            burst_size: <string>
      interfaces: # optional
        <interface_name>:
          ingress_policy: <policy_name> # optional
          egress_policy: <policy_name> # optional

# --- build_prefix_list ---
- type: build_prefix_list
  selector: <cel_selector>
  prefix_list_name: <string>
  entries:
    - prefix: <cidr>
      action: permit | deny
      le: <integer> # optional
      ge: <integer> # optional

# --- build_community_list ---
- type: build_community_list
  selector: <cel_selector>
  community_list_name: <string>
  entries:
    - community: <string> # e.g. "65001:1000"
      action: permit | deny

# --- generate_safe_bgp_filters ---
- type: generate_safe_bgp_filters
  selector: <cel_selector>
  prefix_list_name: <string> # optional
  policy_name: <string> # optional
  block_rfc1918: <bool> # default: true
  permit_default: <bool> # default: false

# --- Overlay tier ---

# --- build_vxlan ---
- type: build_vxlan
  selector: <cel_selector>
  vni_base: <u32>
  mcast_group_base: <string> # optional

# --- build_evpn ---

```

```

- type: build_evpn
  selector: <cel_selector>
  route_distinguisher_base: <string>
  route_target_base: <string>

# --- Meta tier ---

# --- tag_nodes ---
- type: tag_nodes
  selector: <cel_selector>
  tags: {<key>: <json_value>}

# --- map_hardware_inventory ---
- type: map_hardware_inventory
  selector: <cel_selector>
  chassis_model: <string>
  slots: {<u32>: <linecard_model>}

# --- assert_state (mid-pipeline checkpoint) ---
- type: assert_state
  name: <string>
  severity: error | warning
  select: <cel_selector>
  check: <assertion_check>          # same types as top-level
  help: <string>                   # optional

# --- conditional ---
- type: conditional
  condition: <cel_expression>
  then_primitives: [<primitive>, ...]
  else_primitives: [<primitive>, ...] # optional

# --- custom_primitive ---
- type: custom_primitive
  name: <string>
  parameters: {<string>: <json_value>}
  primitives: [<primitive>, ...]

# --- inject_secrets ---
- type: inject_secrets
  selector: <cel_selector>
  secrets: {<metadata_key>: <env_var_name>}

# --- wasm_plugin (requires --features wasm) ---
- type: wasm_plugin
  selector: <cel_selector>
  plugin_path: <filesystem_path>    # path to .wasm binary

```

assertions:

```

- name: <string>
  severity: error | warning
  select: <cel_selector>
  help: <string>          # optional remediation hint
  check:
    type: field_exists
    field: <field_name>
    # OR
    type: min_count
    count: <integer>
    # OR
    type: max_count
    count: <integer>
    # OR
    type: min_edges
    count: <integer>
    # OR
    type: unique_per_group
    field: <field_name>
    group_by: <field_name>

```

```

# OR
type: field_in_cidr
field: <field_name>
cidr: <cidr>
# OR
type: custom_cel
expression: <cel_expression>
# OR
type: use_rule
name: <rule_name> # references rules: section
# OR
type: is_connected # no additional fields
# OR
type: is_acyclic # no additional fields
# OR
type: is_bipartite # no additional fields
# OR
type: reachability
target_selector: <cel_selector>
# OR
type: match_schema
schema: <json_schema_object>
# OR
type: zone_membership
zone: <zone_name or "any">
min_trust_level: <i32> # optional
# OR
type: zone_policy_exists
from_zone: <zone_name>
to_zone: <zone_name>
expected_action: <string> # optional
# OR
type: no_shadowed_rules # no additional fields
# OR
type: zone_pairs_covered # no additional fields
# OR
type: zone_policy_permits
from_zone: <zone_name>
to_zone: <zone_name>
source: <cidr> # optional
destination: <cidr> # optional
protocol: <string> # optional
destination_port: <u16> # optional
icmp_type: <string> # optional
# OR
type: zone_policy_denies # same fields as above
from_zone: <zone_name>
to_zone: <zone_name>

# -- Blast radius policies --
blast_radius:
- name: <string>
  select: <cel_selector> # optional, defaults to all
  check:
    type: max_deletions
    count: <integer>
    # OR
    type: max_modifications
    count: <integer>
    # OR
    type: max_percentage_changed
    value: <float> # 0.0 to 100.0

# -- Target-specific transforms --
transforms:
- name: <string>
  when: <cel_expression> # optional device predicate
  rules:
    - match_expr: <cel_expression> # selects stanzas

```

```

    apply:
      <field>: <cel_expression>      # computes new field value

# -- Hardware library --
hardware_library:
  chassis:
    - model: <string>
      slots: <integer>
      description: <string>
  linecards:
    - model: <string>
      ports: <integer>
      speed: <string>

```

B Selector DSL Grammar

Listing 14. Selector DSL grammar (informal EBNF).

```

selector    ::= "all()" | target "[" filter "]"
target      ::= "nodes" | "edges"
filter      ::= cel_expression

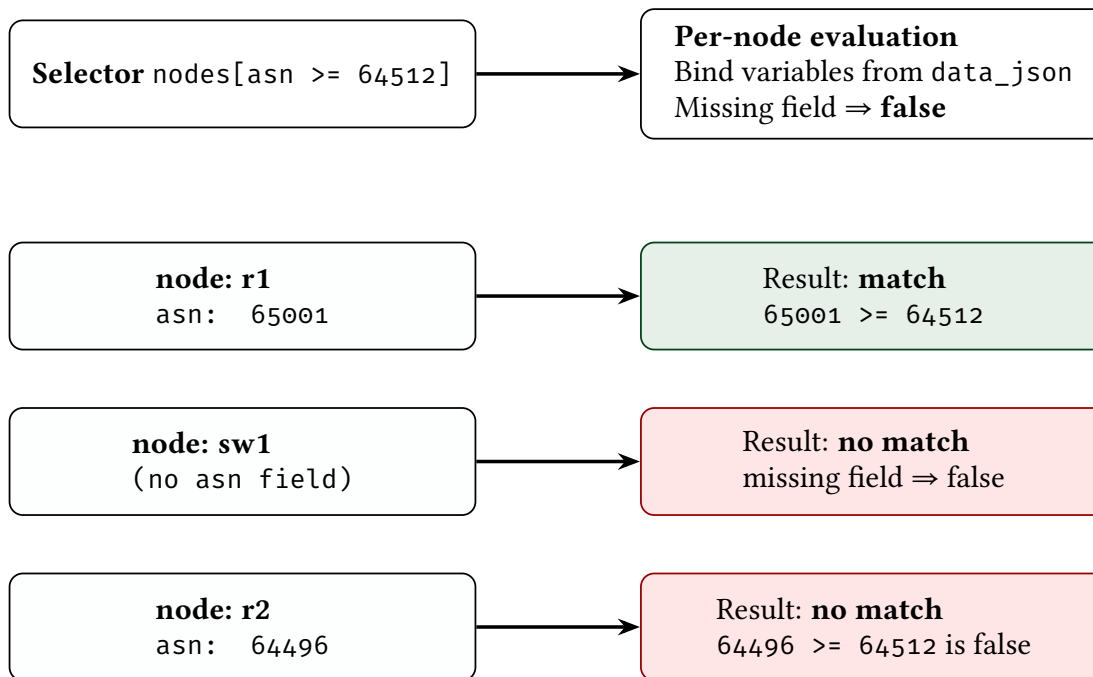
(* Surface syntax normalisations applied before CEL compilation *)
"="         -> "=="
"and"       -> "&&"
"or"        -> "||"
"not"       -> "!"
"&"        -> "&&"
"|"        -> "||"

(* CEL variables bound per entity *)
nodes: id (int), type (string), layer (string), + all data_json fields
edges: id (string), src (int), dst (int), layer (string),
      + all data_json fields

(* IP prefix length extraction *)
For any field with an IP/CIDR value "10.0.0.0/30":
  {field}_len = 30 (integer, bound automatically)

(* Undeclared references *)
Accessing a field that does not exist on a node evaluates to false
(not an error), enabling heterogeneous topologies.

```



This design makes selectors robust on mixed device roles: switches can omit routing fields without causing compilation errors or special-case logic.

Figure 34. Selector evaluation on heterogeneous nodes. Missing fields do not raise errors; they short-circuit to false, enabling a single selector to be applied across mixed node types without explicit guards.

C CLI Quick Reference

Table 41. CLI command summary.

Command	Synopsis
generate	Full pipeline: parse → transform → render → write
validate	Dry-run pipeline with optional template pre-flight
plan	Compute and display topology mutation plan (or config diff)
inspect	AST visualisation, Graphviz DOT export, or per-node trace
ci-run	Combined validate + plan + config diff for CI/CD
import	Import YAML topology to NTE archive
lint	Run design-rule assertions only
fmt	Standardise blueprint YAML formatting
impact	Calculate blast radius of a blueprint change
report	Generate Markdown architecture report
export-clab	Generate Containerlab simulation file
sync	Synchronise topology with NSoT (pull/push)

List of Figures

- 1 Conceptual compilation funnel. The primary input is the annotated topology; the blueprint selects compilation strategies; the compiler derives an enriched layered graph; the backend lowers that graph into vendor-neutral and then target-specific ASTs before serialisation. 3
- 2 The layered graph model. Physical devices (bottom) have corresponding protocol nodes (top). Dashed arrows show the parent mapping (`_source_id`) that traces each protocol node back to its physical origin. Protocol-layer edges represent logical intent relationships and may be derived from, but need not mirror, physical links. 4

3	Layered stack as bands. Each column is a physical device (A, B, C). Each protocol layer adds one node per participating device, linked back to the physical node via the parent mapping (vertical dashed arrows). Overlay edges encode protocol-specific logical relationships and may differ across layers.	5
4	Cross-layer state resolution. Protocol overlays do not maintain redundant state. They resolve physical facts (like allocated IPs) on-the-fly by querying their host physical node via the <code>_source_id</code> parent mapping.	5
5	Target backend package structure. Templates render per-vendor syntax; <code>transforms.yaml</code> declares TSDM lowering rules; <code>hardware-inventory/</code> maps chassis models to slot layouts. Most organisations author this once per platform and reuse it across sites.	8
6	Progressive model enrichment. Left: <code>mesh_nodes</code> establishes edges between bare nodes. Centre: <code>provision_ips</code> annotates nodes with IP addresses. Right: <code>build_protocol_layer</code> clones nodes into a protocol layer and deep-merges a protocol overlay into the clones; dashed arrows show the parent mapping back to the physical devices. At each stage the model grows richer; no primitive needs to know what others have done.	9
7	The select-enrich-validate cycle. Every primitive selects a population of nodes or edges and enriches them with new state. Validation is expressed as assertions over the model and may run as an explicit checkpoint, a layer barrier, or a final gate before rendering.	10
8	Selector visibility timeline. Each primitive's selector evaluates against the <i>current</i> graph state, which includes all fields written by earlier primitives. State accumulates monotonically.	11
9	DeviceContext as the compiler's per-node state carrier. Primitives enrich metadata and typed models; mapping emits structured stanzas; TSDM rewrites those stanzas; templates then serialise the final per-node state without introducing new semantics.	12
10	From whiteboard to compiled model. An architect's visual intent (left, optional) informs both the annotated topology (the network's "source code": nodes, roles, sites, properties) and the blueprint (which compilation strategies to apply). Together, these feed the compiler, which produces an enriched, layered graph (right).	13
11	Policy compilation flow. The operator declares zone intent; the <code>build_zone_policy</code> primitive compiles it into typed rule objects on DeviceContext; vendor templates render platform-specific syntax.	14
12	Selector-to-annotation data flow. Topology annotations are the values that selectors query; the selector filters nodes into a matched set; the primitive then operates on that set. The topology annotations are the "input language" and the selector is the bridge between topology intent and blueprint strategy.	16
13	Primitive tier hierarchy. Lower tiers create structure; upper tiers annotate and configure existing nodes. Each tier builds on state written by earlier tiers.	18
14	Visual intuition for <code>no_shadowed_rules</code> . The assertion flags rules whose match space is completely covered by earlier rules, making them unreachable (and often indicating operator intent errors).	27
15	Assertion firing points. CP : mid-pipeline checkpoint (inline <code>assert_state</code>); LB : layer barrier (runs after a layer's primitives complete); FG : final gate (runs before rendering). All three evaluate the same assertion types but fire at different pipeline positions for earlier feedback.	27
16	NetCfg compilation pipeline. High-level intent is progressively lowered through three distinct AST representations (Verified Graph, Neutral AST, Concrete AST) before serialisation.	29
17	Transactional Safety via the Shadow Topology. Mutations occur on an isolated clone. If any primitive or assertion fails, the shadow is discarded without side-effects on the production graph.	29
18	Mapping evaluation. Each rule's selector matches a subset of the enriched graph; the stanza template extracts fields into a structured DeviceIR unit attached to the node's DeviceContext.	30
19	AST-to-AST lowering via TSDM. Vendor-neutral DeviceIR is transformed into a target-specific data model using rules evaluated against the device's metadata.	30
20	TSDM as a rewrite system. Rules match stanzas structurally and apply deterministic field mutations, producing a concrete AST suitable for vendor-specific rendering without embedding hardware logic in templates.	30
21	TSDM lowering pipeline. DeviceIR stanzas pass through bridge translation, optional IPv6 injection, target-specific AST rewriting, template rendering, and transactional file output.	31
22	Template rendering pipeline. DeviceIR stanzas are resolved to a vendor template, combined with a render context, and emitted by the MiniJinja engine as configuration text plus audit artefacts.	31

23	Determinism as a set of enforced constraints. Each common source of nondeterminism is eliminated by canonical ordering, semantic keys, transactional isolation, and keeping the render layer free of semantic logic.	31
24	Five-pass IPAM algorithm. Passes 1–3 gather state and reserve existing allocations; pass 4 allocates new subnets; pass 5 writes results to DeviceContext. The two-phase structure (read then write) satisfies Rust’s ownership rules without unsafe code.	32
25	Allocator state model. Allocations are partitioned by VRF and keyed by a stable semantic edge identity (hostnames, not integer IDs). Existing allocations are pre-reserved for stickiness; new prefixes are chosen from the free space within the VRF namespace, then written back to node DeviceContext.	32
26	Hierarchical IP pool carving with VRF isolation.	33
27	MLAG/vPC topology mapping. Two physical nodes collapse into a single logical protocol node via array_source_id.	34
28	Protocol overlay construction as a clone + deep merge. The cloned node inherits prior enrichment (e.g., IPs) from the physical node and overlays protocol-specific configuration; the _source_id link provides cross-layer traceability without cross-layer edges.	34
29	Diff engine flow. Two topologies are decomposed by layer, diffed by semantic key, and merged into a deterministic mutation plan containing six change types.	35
30	Why semantic keys matter. Matching by integer ID turns benign insertions into widespread “updates”; matching by a composite semantic key isolates changes to the nodes that actually changed.	35
31	Diffing as semantic key alignment. Nodes are matched by composite meaning (layer, hostname, protocol_type), then compared field-by-field; unmatched keys yield clean add/remove operations instead of cascaded updates.	35
32	The NetDevOps CI/CD pipeline enabled by ci-run.	38
33	NSoT synchronisation data flow.	40
34	Selector evaluation on heterogeneous nodes. Missing fields do not raise errors; they short-circuit to false, enabling a single selector to be applied across mixed node types without explicit guards.	56

List of Tables

1	Stage outputs (conceptual artefacts). These conceptual stages map to the five pipeline layers in §6: Compose = Layer 1, Lower = Layer 3, Adapt = Layer 4, Serialise = Layer 5.	4
2	Well-known topology annotation keys. Keys marked with * are computed by the selector engine and need not appear in the topology file. All other keys are authored by the operator (or imported via netcfg sync pull).	7
3	Per-node device context: fields populated by the compiler pipeline. Fields marked with † are written by specific primitives; all others originate from the annotated topology.	12
4	The five concerns of a network blueprint.	13
5	Primitive canonical names and DSL aliases.	18
6	Primitive catalogue overview. Status: Stable , Beta , Experimental	19
7	mesh_nodes fields.	19
8	provision_ips fields.	20
9	build_protocol_layer fields.	20
10	allocate_resources fields.	20
11	global_pool fields.	20
12	global_resource_pool fields.	21
13	conditional fields.	21
14	custom_primitive fields.	21
15	inject_secrets fields.	21
16	build_routing_policy fields.	21
17	build_access_policy fields.	22
18	wasm_plugin fields.	22
19	build_prefix_list fields.	23
20	build_community_list fields.	23
21	generate_safe_bgp_filters fields.	23

22	<code>build_zone_policy</code> fields.	24
23	<code>build_nat_policy</code> fields.	24
24	NAT rule match fields (<code>NatMatchSpec</code>).	24
25	<code>build_qos_policy</code> fields.	24
26	<code>build_vxlan</code> fields.	25
27	<code>build_evpn</code> fields.	25
28	<code>tag_nodes</code> fields.	25
29	<code>map_hardware_inventory</code> fields.	26
30	<code>assert_state</code> fields.	26
31	Assertion check types.	27
32	Blast radius check types.	28
33	<code>BlastRadiusSpec</code> fields.	28
34	Transform rewrite rule fields.	28
35	CLI commands and the pipeline layers they invoke. ✓ = executes; ◦ = optional (via flag).	37
36	Assertions in the financial compliance case study.	43
37	Error enums and their variants.	43
38	Compile endpoint request and response schema.	44
39	Case study corpus. Each blueprint was compiled successfully with all assertions passing. ...	45
40	Micro-benchmark results (median, Criterion).	45
41	CLI command summary.	56

List of Design Decisions & Rules

1	Design Rule: Hardware Independence	2
2	Design Rule: Node-Centric State	13

Revision History

Version	Date	Changes
2.3.0	March 2026	Ph.D. Insights & Design Rules; v2.3 Standard.
1.1.0	March 2026	sk-techreport v2.2; Synthesis Funnel.
1.0.0	Jan 2026	Initial release of the Rust-based configuration compiler.